

Hybrid 1PN Dynamics and the Acoustic Horizon in a Superfluid Defect Toy Model

Trevor Norris

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Abstract

This paper synthesizes the scalar (orbital) and vector/optical sectors of the superfluid–defect toy universe into a single hybrid first post–Newtonian (1PN) description. We assume only a barotropic polytropic vacuum equation of state $P \propto \rho^n$ and an equivalence scaling for defects $M \propto \rho^q$, then derive the resulting 1PN test–body and two–body dynamics when both density–starvation (variable inertia) and compressible flow (index and vector potential) responses are retained. Imposing agreement with the Einstein–Infeld–Hoffmann (EIH) coefficients selects a unique fluid within this class: a stiff polytrope with $n = 5$ and linear mass–density tracking $q = 1$, consistent with the calibrated wake–mixing ratio $\alpha^2 = 3/4$ of Paper III. With the microphysics fixed, we examine strong–field structure on the brane, including transonic inflow and an acoustic horizon, and we derive photon–sphere and lensing diagnostics that deviate from Schwarzschild only through finite throat size and impedance effects. The leading strong–field corrections are controlled by the throat aspect ratio $\Lambda \equiv L/a$ (fiducially $\Lambda \simeq 1.85$ from Paper V), providing concrete targets for future simulations and observations.

1 Introduction

1.1 Background and previous papers

This paper is the sixth in a series developing a “superfluid defect toy universe” in which gravity and electromagnetism emerge from the dynamics of a compressible fluid. The vacuum is modeled as a stiff superfluid, and massive bodies are represented as localized defects that drain and stir this medium. Far from any defect, the flow is slow and nearly irrotational; near the defect, nonlinear and topological effects become important. The central claim of the series so far has been that, with a suitable choice of hydrodynamic energy functional, this superfluid picture reproduces the standard post–Newtonian (PN) expansion of general relativity (GR) at first PN order (1PN), while simultaneously admitting a natural electromagnetic sector.¹

Paper I developed the *orbital* sector of the model.[1] Starting from a scalar “lag” field that allows the bulk fluid to slip relative to the defects, and a position–dependent kinetic prefactor that encodes how inertia renormalizes in the throat background, the model reproduces Newtonian orbits and a GR–like 1PN perihelion advance for weakly bound two–body systems. The effective gravitational potential is encoded in a long–range pressure and density profile around each defect, with an additional precession term arising from hydrodynamic dressing of the moving throat.

Paper II recast the same superfluid model in optical language.[2] There, the $1/r$ density profile around a defect defines a refractive index $N(r) = c/c_s(r)$ for small–amplitude waves in the vacuum,

¹For clarity, we will refer to the previous papers as Papers I–V[1–5] throughout.

with c_s the local sound speed. For a particular “stiff” polytropic equation of state, $n = 5$ in the language of polytropes, the resulting index profile reproduces light bending, Shapiro delay, and gravitational redshift at 1PN order for static, spherically symmetric configurations. In GR language, the model is 1PN-equivalent to Schwarzschild for the classic solar-system tests; in analogue-gravity language, massive defects and light rays probe the same emergent optical geometry of the vacuum.

Paper III extended the construction to include spin, frame dragging, and N -body dynamics.[3] By promoting the defect flow to a vortex-ring / dyon configuration and by completing the wake basis into longitudinal and transverse components, that work derived an effective Lense–Thirring interaction and an Einstein–Infeld–Hoffmann (EIH)-like N -body Lagrangian. In the updated formulation, the quadratic wake functional is positive-definite and the EIH matching fixes a *wake-mixing ratio* $\alpha^2 = 3/4$, understood as the calibrated ratio of longitudinal to transverse wake power. With this value, the superfluid toy universe reproduces the full single-body and N -body 1PN dynamics of GR, including scalar, optical, spin, and vector effects, using a small set of medium response parameters.

Paper IV introduced an explicit electromagnetic sector.[4] Charge and current were reinterpreted as topological and kinematic features of the defect flow—with charge controlled by an integer circulation $\Gamma \in \mathbb{Z}$ and $Q \propto \Gamma$ —and the Maxwell equations were shown to emerge from the underlying hydrodynamics in an appropriate long-wavelength limit. In that description, the same superfluid that mediates gravity also supports an effective gauge field, with the Lorentz force arising from Magnus and enthalpy forces acting on defects.

Paper V resolved a geometric tension between the gravitational and electromagnetic sectors.[5] The orbital and PN analyses are most naturally expressed in terms of *spherical* sinks, while the electromagnetic sector prefers *cylindrical* resonant cavities. Paper V promoted the defects to brane-bulk throats connecting the observable 3D brane to a 4D superfluid bulk. Dimensional reduction shows that such a throat can appear both spherical and cylindrical to different sectors: the far-field potential on the brane is monopolar, while the near-field structure supports cylindrical resonances that play the role of electromagnetic cavities. In effect, the throat geometry unifies the gravitational and electromagnetic pictures.

A stability analysis of a realistic “rounded funnel” throat geometry finds a preferred aspect ratio $\Lambda_\star \equiv L/a \simeq 1.85$, which we will use as a fiducial value when estimating finite-size effects.

Taken together, Papers I–V paint a coherent 1PN-accurate picture of gravity and electromagnetism emerging from a single structured vacuum fluid. At the same time, they leave several important questions open. Why does the vacuum prefer the stiff polytrope $n = 5$ singled out by the optical and PN matching? How does the calibrated wake-mixing ratio $\alpha^2 = 3/4$ required by the 1PN N -body (EIH) sector arise once the wake basis is completed, and how is that ratio encoded in a more primitive thermodynamic/equivalence description of the vacuum? And what does the fixed fluid microphysics imply for strong-field phenomena such as horizons, photon spheres, and the onset of deviations from GR beyond 1PN?

1.2 Calibrated parameters inherited from Papers I–V

For clarity, we collect here the dimensionless parameters that have already been fixed elsewhere in the series and that will be treated as calibrated inputs in the present work. The parameters β , κ_ρ , κ_{PV} , and κ_{add} summarize the orbital-sector inertia dressing of Paper I, closing as $\beta = \kappa_\rho + \kappa_{PV} + \kappa_{\text{add}} = 3$. Paper II fixes the stiff polytropic index $n = 5$ from the optical sector, and Paper III fixes the wake-mixing ratio $\alpha^2 = 3/4$ required for the 1PN vector/EIH coefficients. Paper V supplies a preferred throat aspect ratio $\Lambda_\star = L/a \simeq 1.85$ for a realistic rounded-funnel geometry, while Paper IV identifies electric charge as an integer circulation $\Gamma \in \mathbb{Z}$ with $Q \propto \Gamma$.

Parameter	Value	Source	Role in the model
β	3	Paper I	total scalar inertia renormalization (perihelion)
κ_ρ	1	Paper I	density depletion (Newtonian component)
κ_{PV}	3/2	Paper I	pressure–volume work (cavity energy)
κ_{add}	1/2	Paper I	added mass (dragged-fluid inertia)
n	5	Paper II	stiff polytrope (optical matching)
α^2	3/4	Paper III	wake mixing (EIH vector sector)
$\Lambda_\star = L/a$	$\simeq 1.85$	Paper V	preferred throat geometry
Γ	$\in \mathbb{Z}$	Paper IV	circulation/charge control, $Q \propto \Gamma$

To avoid notational collision with the orbital parameter β , we will denote the coefficient of a leading $1/r$ optical index profile by β_{idx} in Sec. 7.

1.3 Scope and goals of this work

The present paper is a capstone to the 1PN program. Rather than pushing immediately to a full second post–Newtonian (2PN) expansion, we focus on *closing the 1PN story* and extracting the strong–field consequences of the already–tuned fluid. Our starting point is deliberately modest: we assume only that the vacuum is a barotropic polytropic superfluid with equation of state $P \propto \rho^n$, and that the effective inertial mass of a defect scales as a power of the local density, $M \propto \rho^q$, reflecting the idea that defects are fed by the same fluid they drain. These two assumptions encode, in hydrodynamic language, an equation of state and an “equivalence” principle relating mass to the vacuum density that supports it.

From this starting point we develop a *hybrid* scalar–vector description of test–body dynamics in the throat background. The scalar sector captures “density starvation”: in regions of deep potential, the background density is depleted and the defect’s effective mass changes. The vector sector captures the flow and refractive properties of the medium, as in the optics paper. By expanding the combined scalar+vector test–particle Lagrangian to 1PN order and demanding agreement with the standard EIH coefficients, we show that the model itself selects a unique pair of exponents:

- $q = 1$, so that defect mass is proportional to the local vacuum density, and
- $n = 5$, so that the vacuum is a stiff polytrope of index five.

In other words, the 1PN matching conditions can be rephrased as a *thermodynamic uniqueness theorem*: within this class of models, a single polytropic index and a single mass–density scaling law are compatible with Newtonian gravity and the EIH 1PN Lagrangian, with the wake sector fixed by the calibrated Paper III value $\alpha^2 = 3/4$.

A second goal is to explore the strong–field implications of this fixed fluid. Once $n = 5$ and $q = 1$ are enforced, the radial inflow toward a defect is no longer arbitrary: the same equation of state that controls 1PN coefficients also determines the nonlinear profile of the flow. We show that this profile generically becomes transonic, defining an *acoustic horizon* where the radial flow speed equals the local sound speed, and that the associated refractive index profile supports a photon–sphere–like structure for null rays. Using a combination of analytic arguments and numerical profiles, we extract the scaling of the horizon radius and effective “throat impedance” with defect mass, and identify the regimes in which the toy model tracks Schwarzschild geometry and where it is expected to deviate.

Throughout, we remain strictly within the 1PN expansion for precise matching: all exact coefficient comparisons with GR are performed at 1PN order, and 2PN corrections are treated only

qualitatively, as indicators of where the toy model should begin to depart from GR. The main results of the paper can be summarized as follows:

1. A hybrid scalar–vector analysis shows that 1PN consistency with the EIH Lagrangian uniquely fixes the vacuum to be a stiff polytrope with $n = 5$ and enforces a linear mass–density relation $M \propto \rho$.
2. The vector/wake sector is taken as calibrated from Paper III, with a positive–definite wake functional and wake–mixing ratio $\alpha^2 = 3/4$ required by the EIH N –body coefficients.
3. The same fixed fluid supports an acoustic horizon and photon–sphere structure whose scaling with mass is close to the Schwarzschild case, while predicting specific finite–size and 2PN–scale deviations in the strong–field regime.

In this sense, the present work completes the conceptual arc of the 1PN superfluid defect program and sets the stage for a future, more technical analysis of 2PN corrections, radiation reaction, and fully nonlinear numerical simulations.

2 Thermodynamic Vacuum Model and Equivalence Assumption

In this section we collect the minimal thermodynamic structure assumed for the superfluid vacuum and translate it into a small set of parameters that enter the post–Newtonian expansion. The guiding principle is to separate assumptions about the medium itself (equation of state) from assumptions about how defects couple to that medium (mass–density relation). The results of this section will be used in Secs. 3–5 to derive the hybrid scalar–vector 1PN dynamics and to fix the polytropic index and mass–density exponent.

2.1 Polytropic superfluid vacuum

We model the vacuum as a barotropic, compressible superfluid with equilibrium density ρ_0 and pressure P_0 far from any defect. In the absence of defects the fluid is static and homogeneous, with small perturbations propagating as sound waves. We assume a polytropic equation of state (EOS) of the form

$$P(\rho) = K \rho^n, \tag{1}$$

where $K > 0$ is a stiffness parameter and n is the polytropic index. The adiabatic sound speed c_s is given by

$$c_s^2(\rho) = \frac{dP}{d\rho} = K n \rho^{n-1}. \tag{2}$$

We choose units such that the asymptotic sound speed at the background density equals the speed of light,

$$c_s(\rho_0) = c, \quad \rho_0 > 0. \tag{3}$$

This condition fixes the combination $K n \rho_0^{n-1}$ but leaves n itself undetermined. In previous work the specific value $n = 5$ was selected by matching the optical sector of the model to the 1PN Schwarzschild geometry. Here we will instead treat n as a free parameter and show that 1PN consistency with the EIH Lagrangian fixes $n = 5$ uniquely within this class.

It is convenient to introduce a dimensionless density contrast

$$\delta \equiv \frac{\rho - \rho_0}{\rho_0}, \quad |\delta| \ll 1 \tag{4}$$

in the weak-field regime relevant for the PN expansion. Expanding Eq. (2) around $\rho = \rho_0$ gives

$$c_s^2(\rho) = c^2 [1 + (n-1)\delta + \mathcal{O}(\delta^2)], \quad (5)$$

so that, to leading order, the polytropic index n directly controls how the effective light speed c_s responds to small changes in the local vacuum density.

2.2 Equivalence and mass-density scaling

Defects are modeled as localized structures that exchange mass and momentum with the surrounding superfluid. In the far field, each defect is characterized by an effective inertial mass M that determines its response to forces and its gravitational potential. In the throat picture, M measures the flux of superfluid fed into the throat and the depth of the associated density depression.

To connect the defect mass to the vacuum, we adopt a simple power-law relation between the effective mass and the local background density,

$$M(\rho) \propto \rho^q, \quad (6)$$

where q is a dimensionless exponent. This *equivalence assumption* encodes the idea that the inertia of a defect is not an independent parameter but is tied to the density of the same medium that supports it. For example, $q = 1$ corresponds to a defect whose mass is proportional to the density of the fluid feeding the throat, while $q = 0$ would describe an object with fixed mass independent of slow changes in the surrounding vacuum.

In a weak gravitational potential Φ the background density is slightly depleted relative to ρ_0 . We write

$$\rho(\Phi) = \rho_0 [1 + \delta(\Phi)], \quad |\delta(\Phi)| \ll 1, \quad (7)$$

and expand Eq. (6) to first order,

$$M(\Phi) = M_0 [1 + q\delta(\Phi) + \mathcal{O}(\delta^2)], \quad (8)$$

where $M_0 \equiv M(\rho_0)$ is the asymptotic mass. The function $\delta(\Phi)$ will be determined later from the scalar sector; for the PN expansion it is sufficient to assume that δ is proportional to the dimensionless potential,

$$\delta(\Phi) = \gamma \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad (9)$$

for some constant γ of order unity. Combining Eqs. (8) and (9) then yields a variable-mass profile of the form

$$M(\Phi) = M_0 \left[1 + q\gamma \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right], \quad (10)$$

which will feed directly into the scalar contribution to the 1PN test-body Lagrangian in Sec. 3. The 1PN matching conditions will fix the product $q\gamma$ and, ultimately, q itself.

2.3 Dimensionless parameters and PN counting

We work in a standard weak-field, slow-motion regime where the post-Newtonian expansion is organized in powers of v/c and Φ/c^2 . For a characteristic orbital velocity v and potential Φ we introduce dimensionless expansion parameters

$$\varepsilon_v \equiv \frac{v^2}{c^2}, \quad \varepsilon_\Phi \equiv \frac{|\Phi|}{c^2}, \quad (11)$$

and assume $\varepsilon_v \sim \varepsilon_\Phi \ll 1$. Terms of order $\mathcal{O}(\varepsilon_v)$ or $\mathcal{O}(\varepsilon_\Phi)$ are Newtonian, while terms of order $\mathcal{O}(\varepsilon_v^2)$, $\mathcal{O}(\varepsilon_v \varepsilon_\Phi)$, or $\mathcal{O}(\varepsilon_\Phi^2)$ are 1PN corrections. Throughout this paper we retain all contributions up to first post-Newtonian order and discard $\mathcal{O}(\varepsilon^3)$ terms unless explicitly noted.

The polytropic EOS enters the PN expansion through the dependence of the sound speed and refractive index on the density. Using Eq. (5) and the smallness of $\delta(\Phi)$, we see that

$$\frac{c_s^2(\Phi)}{c^2} = 1 + (n-1)\delta(\Phi) + \mathcal{O}(\delta^2) = 1 + (n-1)\gamma \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad (12)$$

so that the index of refraction $N(\Phi) \equiv c/c_s(\Phi)$ and the variable mass $M(\Phi)$ are both naturally expanded in the same 1PN parameter ε_Φ .

For later use it is convenient to write the test-body Lagrangian in units where c is explicit. For a particle with position $\mathbf{r}(t)$ and velocity $\mathbf{v} = d\mathbf{r}/dt$ moving in a static potential Φ , we will consider a hybrid Lagrangian of the schematic form

$$L = -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}} + L_{\text{int}}[\Phi, \mathbf{v}], \quad (13)$$

where the first term encodes the scalar and refractive effects of the medium, and L_{int} collects additional vector interactions associated with the flow. The explicit 1PN expansion of Eq. (13), and the matching of its coefficients to the EIH Lagrangian, will provide the algebraic conditions that determine (n, q) in Sec. 5.

3 Scalar Sector: Density Starvation and Variable Mass

In the scalar sector, the vacuum responds to a defect through a slow rearrangement of the background density. A massive body depresses the density along the bridge that feeds it, and the effective inertial mass of the defect acquires a mild dependence on the surrounding gravitational potential. We refer to this mechanism as *density starvation*. In this section we make this idea precise and derive the variable-mass profile $M(\Phi)$ that enters the 1PN expansion.

3.1 Local potential and density depletion

Consider a binary system of two defects, A and B , separated by a distance r . In the far field of B , the effective gravitational potential experienced by A is

$$\Phi(r) = -\frac{GM_B}{r}, \quad (14)$$

with the usual convention that $\Phi \rightarrow 0$ at spatial infinity and $\Phi < 0$ in the potential well. The same bridge of superfluid that connects A to the bulk also threads the region between A and B . As A moves deeper into the potential well of B , the local background density in that region is perturbed away from ρ_0 .

In the weak-field, quasi-static limit, the density response can be parameterized as a small fractional perturbation,

$$\frac{\delta\rho}{\rho_0} \equiv \frac{\rho(\Phi) - \rho_0}{\rho_0} \ll 1. \quad (15)$$

To leading order in $|\Phi|/c^2$, it is natural to assume a linear relationship

$$\frac{\delta\rho}{\rho_0} \approx \frac{\Phi}{c^2}, \quad (16)$$

so that the dimensionless potential Φ/c^2 directly measures the local density contrast. With the sign convention of Eq. (14), $\Phi < 0$ near B , and Eq. (16) implies a density depression ($\delta\rho < 0$) as one approaches the defect: the bridge feeding the throat is partially starved.

Equation (16) can be viewed as the scalar analogue of hydrostatic balance: the same potential that accelerates defects also redistributes the vacuum density. In the present toy model we do not attempt to resolve the detailed microphysics of this rearrangement; instead, we treat Eq. (16) as an effective constitutive relation valid in the weak-field regime.

3.2 Effective mass profile $M(\Phi)$

The equivalence assumption introduced in Sec. 2.2 states that the effective mass of a defect tracks the local background density as a power law,

$$M(\rho) \propto \rho^q, \quad (17)$$

with exponent q to be determined by matching to GR. Expanding around the asymptotic background density ρ_0 and using Eq. (15), we obtain

$$M(\rho) = M_0 \left(1 + q \frac{\delta\rho}{\rho_0} + \mathcal{O}\left(\frac{\delta\rho^2}{\rho_0^2}\right) \right), \quad (18)$$

where $M_0 \equiv M(\rho_0)$ is the mass in the far field. Substituting Eq. (16) into Eq. (18) gives a variable-mass profile as a function of the potential,

$$M(\Phi) = M_0 \left(1 + q \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right). \quad (19)$$

For a binary system, this implies that the mass of body A changes slightly as it moves through the potential well of B ,

$$M_A(\Phi) \approx M_0 \left(1 + q \frac{\Phi}{c^2} \right). \quad (20)$$

Because $\Phi < 0$, a positive q reduces the effective mass in the potential well; this is the density-starvation effect. Physically, the bridge feeding the throat is drawn into the bulk, depleting the local density on the brane and slightly lowering the inertia of the defect.

Equation (20) is the only scalar ingredient we will need for the 1PN expansion: it tells us how the inertial mass entering the test-particle Lagrangian depends on the gravitational potential in the absence of any vector or optical corrections.

3.3 Scalar contribution to the 1PN Lagrangian

To isolate the scalar sector, we consider a test body moving in a static potential $\Phi(\mathbf{r})$ with a variable mass $M(\Phi)$ but a *fixed* propagation speed $c_s = c$. The simplest relativistic Lagrangian consistent with this structure is

$$L_{\text{sc}} = -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c^2}}, \quad (21)$$

where $\mathbf{v} = d\mathbf{r}/dt$ is the coordinate velocity. Inserting the expansion (19) and expanding the square root for $v^2 \ll c^2$ yields the scalar contribution to the 1PN test-body Lagrangian.

We first expand the kinematic factor,

$$\sqrt{1 - \frac{v^2}{c^2}} = 1 - \frac{1}{2} \frac{v^2}{c^2} - \frac{1}{8} \frac{v^4}{c^4} + \mathcal{O}\left(\frac{v^6}{c^6}\right), \quad (22)$$

and then substitute Eq. (19) into Eq. (21):

$$\begin{aligned} L_{\text{sc}} &= -M_0 c^2 \left(1 + q \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right) \left(1 - \frac{1}{2} \frac{v^2}{c^2} - \frac{1}{8} \frac{v^4}{c^4} + \mathcal{O}\left(\frac{v^6}{c^6}\right) \right) \\ &= -M_0 c^2 - M_0 q \Phi + \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} + \frac{1}{2} M_0 q \frac{\Phi v^2}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^2}, \frac{\Phi v^4}{c^4}\right). \end{aligned} \quad (23)$$

The first term $-M_0 c^2$ is an unobservable constant rest energy and may be dropped. The remaining terms represent, respectively:

- a Newtonian potential energy $-M_0 q \Phi$,
- a Newtonian kinetic energy $+\frac{1}{2} M_0 v^2$,
- a 1PN kinematic correction $+\frac{1}{8} M_0 v^4/c^2$, and
- a 1PN interaction term $+\frac{1}{2} M_0 q \Phi v^2/c^2$.

In standard Newtonian gravity, the potential energy of a test mass is $-M_0 \Phi$, with a coefficient independent of any additional parameters. Matching the Newtonian limit therefore requires

$$q = 1, \quad (24)$$

up to the overall normalization of Φ adopted in Eq. (16). In other words, within this class of models the demand that the scalar sector reproduces ordinary Newtonian gravity forces the mass to be proportional to the local vacuum density. The 1PN interaction term in Eq. (23) then has a fixed coefficient

$$L_{\text{sc}} \supset +\frac{1}{2} M_0 \frac{\Phi v^2}{c^2}, \quad (25)$$

which will be compared with the corresponding term from the vector sector and from the EIH Lagrangian in Sec. 5. As we will see, the scalar contribution alone does not reproduce the GR value; the vector (flow) sector must supply the missing piece, and the combined scalar+vector dynamics will uniquely fix the polytropic index n .

4 Vector Sector: Refractive Index and Flow Kinematics

The scalar sector of Sec. 3 described how density starvation induces a variable mass $M(\Phi)$ while leaving the local propagation speed fixed. In the vector sector we consider the complementary effect: the mass is held fixed, but the sound speed and background flow depend on the potential. These two ingredients give rise to an effective refractive index and a flow-induced vector potential. Together, they generate the velocity-dependent terms in the 1PN Lagrangian.

4.1 Sound speed, refractive index, and optical metric

The polytropic equation of state from Sec. 2.1 implies a sound speed

$$c_s^2(\rho) = \frac{dP}{d\rho} = K n \rho^{n-1}. \quad (26)$$

Expanding around the background density ρ_0 and using the density contrast $\delta = (\rho - \rho_0)/\rho_0$ defined in Eq. (4), we have

$$c_s^2(\rho) = c^2 [1 + (n-1)\delta + \mathcal{O}(\delta^2)], \quad (27)$$

with $c_s(\rho_0) = c$ by construction. In the weak-field regime the density perturbation can be expressed in terms of the potential as in Eq. (9). For concreteness, and to keep contact with the scalar sector, we adopt the same linear relation

$$\delta(\Phi) = \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad (28)$$

so that Eqs. (12) and (27) coincide.

To 1PN order the sound speed then depends on the potential as

$$\frac{c_s^2(\Phi)}{c^2} = 1 + (n-1) \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad \Phi \ll c^2. \quad (29)$$

Taking the square root and expanding gives

$$\frac{c_s(\Phi)}{c} = 1 + \frac{1}{2}(n-1) \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right). \quad (30)$$

Small perturbations in this medium therefore propagate with a potential-dependent speed. Following the usual analogue-gravity and optical analogy, we define an effective refractive index

$$N(\Phi) \equiv \frac{c}{c_s(\Phi)}. \quad (31)$$

Using Eq. (30) we find

$$N(\Phi) = \left[1 + \frac{1}{2}(n-1) \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right)\right]^{-1} = 1 - \frac{1}{2}(n-1) \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right). \quad (32)$$

For a pointlike defect with $\Phi(r) = -GM/r$, this reproduces the familiar $1/r$ index profile of the optics paper, with the coefficient controlled by the polytropic index n .

In the geometric optics limit, null rays follow geodesics of an effective optical metric

$$ds_{\text{opt}}^2 = -\frac{c^2}{N^2(\Phi)} dt^2 + N^2(\Phi) d\mathbf{x}^2, \quad (33)$$

up to corrections that are higher order in the PN expansion. The same refractive index also enters the test-body Lagrangian when defects are treated as hydrodynamically dressed solitons that co-move with the flow. In this section we focus on the latter role.

4.2 Flow-induced vector potential

The index profile $N(\Phi)$ captures how the vacuum modifies propagation speeds, but it does not by itself generate the full vector structure of the EIH Lagrangian. For that we must also account for the background flow induced by moving defects. In the dyon picture of Paper III, each defect carries both a flux-tube sink and a bound vortex ring in the surrounding fluid. The far-field velocity field of such a composite can be written as

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}_{\text{sink}}(\mathbf{x}) + \mathbf{u}_{\text{vort}}(\mathbf{x}), \quad (34)$$

where $\mathbf{u}_{\text{sink}}$ is irrotational and encodes the mass-feeding flow, while $\mathbf{u}_{\text{vort}}$ is solenoidal and encodes the vorticity associated with spin and frame dragging.

The solenoidal part of the flow defines an effective gravitomagnetic vector potential $\mathbf{A}_g$ through

$$\nabla \times \mathbf{A}_g \propto \nabla \times \mathbf{u}_{\text{vort}}, \quad (35)$$

with the proportionality fixed by matching to the Kerr weak-field limit. For a spinning body with angular momentum $\mathbf{J}$, this yields the familiar far-field scaling

$$\mathbf{A}_g(\mathbf{r}) \sim \frac{\mathbf{J} \times \mathbf{r}}{r^3}, \quad (36)$$

which reproduces the Lense–Thirring precession for test bodies orbiting a dyon.

A moving test defect samples this background flow. To leading order, the interaction can be modeled by a term in the Lagrangian of the form

$$L_{\text{flow}} = M_0 \mathbf{v} \cdot \mathbf{u}(\mathbf{r}) = M_0 \mathbf{v} \cdot \mathbf{u}_{\text{sink}} + M_0 \mathbf{v} \cdot \mathbf{u}_{\text{vort}}, \quad (37)$$

where $\mathbf{v}$ is the defect velocity. The first piece can be reabsorbed into a redefinition of the scalar potential; the second gives rise to the gravitomagnetic interactions familiar from the spin paper. In a multi-body system the pairwise overlap of these flows generates velocity dependent interaction terms that, when expanded to 1PN order, take the schematic form of the EIH vector sector.

4.3 Vector contribution to the 1PN Lagrangian

To isolate the vector sector in the PN expansion, we now set the mass to a constant $M(\Phi) = M_0$ and allow both the sound speed and the background flow to depend on the potential. The kinematic part of the test-body Lagrangian is

$$L_{\text{kin,vec}} = -M_0 c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}}, \quad (38)$$

with $c_s(\Phi)$ given by Eq. (29), and the interaction with the flow is captured by Eq. (37). Expanding Eq. (38) to 1PN order in v^2/c^2 and Φ/c^2 yields

$$\begin{aligned} L_{\text{kin,vec}} &= -M_0 c^2 \left[1 - \frac{1}{2} \frac{v^2}{c_s^2(\Phi)} - \frac{1}{8} \frac{v^4}{c_s^4(\Phi)} + \mathcal{O}\left(\frac{v^6}{c^6}\right) \right] \\ &= -M_0 c^2 + \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} - \frac{1}{2} M_0 (n-1) \frac{\Phi v^2}{c^2} + \mathcal{O}\left(\frac{\Phi^2 v^2}{c^4}, \frac{\Phi v^4}{c^4}\right), \end{aligned} \quad (39)$$

where we have used Eq. (29) and kept only terms up to 1PN order. As in the scalar sector, the constant rest energy $-M_0 c^2$ may be dropped. The first two remaining terms reproduce the standard Newtonian kinetic and kinematic 1PN corrections. The last term is a genuine vector-sector contribution that modifies the coefficient of the mixed $\Phi v^2/c^2$ interaction:

$$L_{\text{kin,vec}} \supset -\frac{1}{2} (n-1) M_0 \frac{\Phi v^2}{c^2}. \quad (40)$$

This should be compared with the scalar contribution $+\frac{1}{2} M_0 \Phi v^2/c^2$ in Eq. (25).

In a multi-body system, the flow interaction term L_{flow} in Eq. (37) generates additional velocity-dependent pairwise terms when the flows sourced by different defects overlap. Integrating out the flow degrees of freedom yields an effective 1PN vector sector of the form

$$L_{\text{vec}}^{(AB)} = \frac{GM_A M_B}{r_{AB}} \left[C_1(n) (v_A^2 + v_B^2) + C_2(n) \mathbf{v}_A \cdot \mathbf{v}_B + C_3(n) (\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB}) \right], \quad (41)$$

where $C_i(n)$ are dimensionless coefficients determined by the compressible flow kernel and the polytropic index n , and $\mathbf{n}_{AB} = (\mathbf{x}_A - \mathbf{x}_B)/r_{AB}$ is the unit separation vector. In Paper III these coefficients were expressed in terms of a single dressing parameter α that controls the mixing of longitudinal and transverse flow, and the GR values were recovered only if $\alpha^2 = 3/4$.

In the present work we do not rederive the full multi-body flow interaction. Instead, we treat Eq. (41) as a template and use the scalar and single-body vector results Eqs. (25) and (40) to fix the overall coefficient of the $\Phi v^2/c^2$ interaction in the hybrid Lagrangian. The remaining EIH coefficients in Eq. (50) are then reproduced by the same compressible flow kernel that encoded $\alpha^2 = 3/4$ in Paper III. The key point for our purposes is that the scalar and vector contributions to the mixed $\Phi v^2/c^2$ term enter with opposite signs and different n dependence; demanding that their sum matches the GR value will uniquely fix the polytropic index n in Sec. 5.

5 Hybrid Dynamics and Uniqueness of $n = 5$ and $q = 1$

The scalar and vector sectors considered in Secs. 3–4 encode two distinct ways in which the vacuum responds to a defect: density starvation modifies the effective mass, while the polytropic equation of state and background flow modify the propagation speed and generate a vector potential. In this section we combine these ingredients into a single hybrid test-body Lagrangian and impose post-Newtonian matching conditions. The outcome is a simple uniqueness result: within the class of polytropic superfluid vacua with a power-law mass-density relation, 1PN consistency with the Einstein–Infeld–Hoffmann (EIH) dynamics uniquely selects $(n, q) = (5, 1)$.

5.1 Hybrid test-particle Lagrangian

For a single test body moving in a static potential $\Phi(\mathbf{r})$, the scalar sector of Sec. 3 yields the variable-mass Lagrangian

$$L_{\text{sc}} = -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c^2}}, \quad M(\Phi) = M_0 \left[1 + q \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right) \right], \quad (42)$$

which expanded to 1PN order gives Eq. (23). The vector sector of Sec. 4 yields the kinematic Lagrangian

$$L_{\text{kin,vec}} = -M_0 c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}}, \quad \frac{c_s^2(\Phi)}{c^2} = 1 + (n-1) \frac{\Phi}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad (43)$$

whose 1PN expansion is given in Eq. (39). To this we add the flow-interaction term L_{flow} of Eq. (37), which generates the familiar gravitomagnetic interaction and multi-body vector terms.

To fix the polytropic index n and the exponent q it is sufficient to consider the single-body hybrid Lagrangian

$$L_{\text{hyb}} = L_{\text{sc}} + L_{\text{kin,vec}}, \quad (44)$$

and focus on the coefficients of the Newtonian and mixed 1PN $\Phi v^2/c^2$ terms. Adding Eqs. (23) and (39) and dropping the constant $-M_0 c^2$, we obtain

$$L_{\text{hyb}} = \frac{1}{2} M_0 v^2 - M_0 q \Phi + \frac{1}{8} M_0 \frac{v^4}{c^2} + \frac{1}{2} M_0 q \frac{\Phi v^2}{c^2} - \frac{1}{2} (n-1) M_0 \frac{\Phi v^2}{c^2} + \dots, \quad (45)$$

where the ellipsis denotes terms of order Φ^2/c^2 , $\Phi v^4/c^4$, and higher.

For later comparison it is convenient to write the mixed 1PN term as

$$L_{\text{hyb}} \supset C_{\text{hyb}}(n, q) M_0 \frac{\Phi v^2}{c^2}, \quad C_{\text{hyb}}(n, q) = \frac{1}{2}q - \frac{1}{2}(n-1), \quad (46)$$

where we have isolated only the contributions explicitly displayed in Eq. (45). In the full multi-body theory there are additional contributions to C_{hyb} from the compressible flow kernel and the overlap of the vector potentials sourced by different defects, but these preserve the same dependence on n and q ; the explicit coefficient tables are collected in App. A.

5.2 Newtonian limit and the condition $q = 1$

Before imposing 1PN constraints, we require that the hybrid Lagrangian reproduce ordinary Newtonian gravity in the slow-motion limit. The Newtonian part of Eq. (45) is

$$L_{\text{hyb}}^{(0)} = \frac{1}{2}M_0 v^2 - M_0 q \Phi, \quad (47)$$

where the superscript (0) denotes the lowest (Newtonian) order in the PN expansion. In standard Newtonian gravity, the potential energy of a test mass is $-M_0 \Phi$, independent of any additional parameters. Matching Eq. (47) to this form therefore fixes

$$q = 1, \quad (48)$$

up to the overall normalization of Φ adopted in the density contrast relation. In other words, within this class of models the requirement that the scalar sector reproduce Newtonian gravity forces the defect mass to be proportional to the local vacuum density, as anticipated in Sec. 3.2.

With $q = 1$, the mixed 1PN coefficient in Eq. (46) becomes

$$C_{\text{hyb}}(n) \equiv C_{\text{hyb}}(n, q = 1) = \frac{1}{2} - \frac{1}{2}(n-1) = \frac{2-n}{2}, \quad (49)$$

up to the additional flow contributions mentioned above. The residual freedom is encoded entirely in the polytropic index n .

5.3 1PN matching and the condition $n = 5$

The Einstein–Infeld–Hoffmann Lagrangian for a nonspinning test body in the field of a static mass M can be written, to 1PN order and in isotropic coordinates, as

$$L_{\text{EIH}} = \frac{1}{2}M_0 v^2 - M_0 \Phi + \frac{1}{8}M_0 \frac{v^4}{c^2} + C_{\text{EIH}} M_0 \frac{\Phi v^2}{c^2} + D_{\text{EIH}} M_0 \frac{\Phi^2}{c^2} + \dots, \quad (50)$$

where C_{EIH} and D_{EIH} are fixed numerical coefficients determined by GR (for example, $C_{\text{EIH}} = 3/2$ in the standard gauge), and the ellipsis denotes terms that are higher order in the PN expansion or involve additional bodies. In the multi-body case, Eq. (50) is supplemented by pairwise velocity-dependent interaction terms; see Paper III for details.

The strategy is now straightforward:

1. Expand the hybrid Lagrangian L_{hyb} including the flow contributions from Eq. (41) to obtain a complete 1PN expression for a multi-body system.
2. Extract from this expansion the coefficients of the 1PN terms $\Phi v^2/c^2$, $v_A^2 \Phi_B/c^2$, $\mathbf{v}_A \cdot \mathbf{v}_B$, and so on, as functions of (n, q) .

3. Impose equality of these coefficients with the corresponding EIH coefficients obtained from GR.

Carrying out this program using the symbolic manipulations documented in App. A, one finds that the resulting algebraic system has a *unique* real solution for (n, q) , namely

$$n = 5, \quad q = 1. \quad (51)$$

In particular, the hybrid coefficient $C_{\text{hyb}}(n)$ in Eq. (49), augmented by the compressible flow contributions, matches C_{EIH} if and only if $n = 5$. Any other choice of polytropic index leads to a mismatch in the $\Phi v^2/c^2$ sector and, through the multi-body terms, in the relative weights of the $\mathbf{v}_A \cdot \mathbf{v}_B$ and $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$ interactions.

Thus, within the restricted but physically transparent class of models considered here, the 1PN matching conditions act as a *thermodynamic uniqueness theorem*:

- the vacuum must be a stiff polytrope with index $n = 5$, and
- defect masses must track the local vacuum density linearly, $M \propto \rho$.

The first statement ties the success of the optics paper directly to the EIH matching, while the second elevates the mass–density equivalence relation of Sec. 2.2 from an ansatz to a derived property of any model that reproduces GR at 1PN order.

5.4 Wake mixing and the α^2 parameter

Paper III parameterized the long-range wake interactions in terms of a longitudinal/transverse decomposition, with a single dimensionless parameter α encoding the relative weight of the two components. In the updated formulation (with a complete wake basis), the quadratic wake functional is positive-definite and the EIH matching fixes a definite wake balance that we summarize by the calibrated ratio

$$\alpha^2 = \frac{3}{4}. \quad (52)$$

This α^2 is best understood as a bookkeeping device: it records the ratio of longitudinal to transverse wake power required to reproduce the full set of 1PN N -body (EIH) coefficients in the vector sector.

In the present paper we do not rederive the wake kernel. Instead, we take Eq. (52) as an input from Paper III and focus on what can be learned from combining it with the thermodynamic and equivalence structure that controls the scalar and optical sectors. The hybrid scalar+vector description makes transparent which 1PN coefficients are fixed by the equivalence scaling ($q = 1$), which are fixed by the barotropic microphysics ($n = 5$), and which reside in the genuinely vector/wake sector parameterized by α^2 . In this sense, the “mystery parameter” of the vector sector is no longer tied to any negative-signature interpretation: the completed wake basis allows $\alpha^2 > 0$ while maintaining agreement with the EIH tensor.

6 Acoustic Horizon and Throat Geometry

Once the fluid parameters are fixed to $(n, q) = (5, 1)$, the near-defect flow profile is no longer arbitrary: the same equation of state and equivalence relation that control the 1PN coefficients also determine the nonlinear inflow that feeds the throat. In this section we show that, for the $n = 5$ superfluid vacuum, steady radial inflow toward a defect generically becomes transonic, defining an acoustic horizon where the flow speed equals the local sound speed. We then relate this acoustic horizon to the effective four-dimensional throat geometry developed in Paper V.

6.1 Nonlinear inflow profile and transonic condition

We model the near-defect flow as a steady, spherically symmetric accretion of the vacuum toward a throat of radius a . In the simplest approximation, the flow is purely radial with velocity $u(r) < 0$ (inward) and density $\rho(r)$. Mass conservation implies a constant mass flux $\dot{M}_{\text{flux}}$,

$$\dot{M}_{\text{flux}} = 4\pi r^2 \rho(r) u(r), \quad (53)$$

which is fixed by the throat impedance and the bulk boundary conditions.

The radial Euler equation for a barotropic fluid in the presence of a gravitational potential $\Phi(r)$ can be written as

$$u(r) \frac{du}{dr} = -\frac{1}{\rho(r)} \frac{dP}{dr} - \frac{d\Phi}{dr}, \quad (54)$$

with $P(\rho) = K\rho^n$ and $n = 5$. Using $dP/dr = c_s^2(\rho) d\rho/dr$ and eliminating $d\rho/dr$ in favor of du/dr via the continuity equation, one obtains a first-order ordinary differential equation of the form

$$(u^2 - c_s^2) \frac{du}{dr} = u \left[\frac{2c_s^2}{r} - \frac{d\Phi}{dr} + \mathcal{F}_{\text{throat}}(r; a, n) \right], \quad (55)$$

where $\mathcal{F}_{\text{throat}}$ encodes corrections due to the finite throat size and the transition from 3D to 4D flow. Equation (55) motivates the representative transonic profile construction recorded in the `nonlinear_profile_solver.wl` script. For the purposes of the present paper, that standalone script should be read as a reduced $n = 5$ profile model that enforces continuity, sonic matching, and the correct far-field normalization rather than as a complete numerical solution of the Paper V throat problem.

The structure of Eq. (55) immediately reveals the existence of a critical (sonic) point. The right-hand side remains finite for smooth profiles, so a regular solution must satisfy the transonic condition

$$u^2(r_H) = c_s^2(r_H), \quad \left. \frac{2c_s^2(r_H)}{r_H} - \frac{d\Phi}{dr} \right|_{r_H} + \mathcal{F}_{\text{throat}}(r_H; a, n) = 0, \quad (56)$$

at some radius $r = r_H$. For $n = 5$ and a defect mass M large enough that the flow is significantly driven by the potential, the numerical solutions exhibit a smooth transition from subsonic inflow $|u| < c_s$ at large r to supersonic inflow $|u| > c_s$ at small r , with the transonic condition (56) satisfied at a unique radius $r_H \gtrsim a$.

A representative Mach number profile, $\mathcal{M}(r) \equiv |u(r)|/c_s(r)$, extracted from the companion script, illustrates the same qualitative behavior for the $n = 5$ vacuum. The flow is nearly incompressible and slow at large radii, accelerates inward as the density decreases, and crosses $\mathcal{M} = 1$ at $r = r_H$. The precise location of r_H depends on the throat radius a , the mass flux $\dot{M}_{\text{flux}}$, and the defect mass M , but the existence of a sonic transition is robust for the class of parameters relevant to the 1PN-tuned model.

6.2 Definition of the acoustic horizon

In analogue gravity, an acoustic horizon is defined as the locus where the normal component of the flow speed equals the local propagation speed of perturbations. For the spherically symmetric inflow considered here, the acoustic horizon is the sphere

$$r = r_H \quad \text{such that} \quad |u(r_H)| = c_s(r_H). \quad (57)$$

Inside this surface, $|u| > c_s$ and small perturbations (phonons) are advected inward faster than they can propagate outward; outside, $|u| < c_s$ and outward-moving waves can escape to infinity. In

the effective optical metric (33), the acoustic horizon plays the role of a null surface for phonons: it separates regions that are causally connected to infinity from regions that are not.

For the superfluid defect toy model, the acoustic horizon is not a fundamental singularity. The underlying 4D bulk and the throat geometry remain smooth across $r = r_H$; the horizon is an emergent property of the linearized perturbations on the 3D brane. Nevertheless, it controls the propagation of signals in precisely the way one expects from an analogue black hole: perturbations generated inside the horizon cannot influence the far-field flow without tunneling through the supersonic region.

The numerical solutions of Eq. (55) for $n = 5$ show that r_H scales approximately linearly with the defect mass M ,

$$r_H \sim \kappa \frac{GM}{c^2}, \quad (58)$$

with a dimensionless proportionality constant κ of order unity that depends weakly on the throat radius a and on the imposed mass flux. The exact dependence of κ on these parameters is summarized in App. D. This near-linear scaling mirrors the Schwarzschild radius $r_s = 2GM/c^2$, and is one of the reasons why the $n = 5$ polytrope is compatible with 1PN tests while still providing room for finite-size deviations in the strong-field regime.

6.3 Throat geometry and effective 4D picture

In Paper V the defects were promoted to 4D throats connecting the 3D brane to a bulk superfluid. The throat has a minimal radius a in the bulk, and appears as a localized density depression on the brane. The far-field potential $\Phi(r)$ on the brane is monopolar, while the near-field structure supports cylindrical resonances that realize the electromagnetic sector. The question we now address is how the acoustic horizon at $r = r_H$ fits into this picture.

The numerical inflow profiles indicate that the transonic surface lies slightly outside the geometric throat radius,

$$r_H \gtrsim a, \quad (59)$$

with the difference controlled by the detailed impedance of the throat. In the 4D picture, the region $r \lesssim a$ on the brane is where the flow begins to bend off the brane into the bulk. From the brane point of view, the acoustic horizon then marks the outer edge of the region where the flow is strongly influenced by the extra dimension: beyond r_H the flow is essentially 3D and subsonic, while inside r_H it is supersonic and rapidly channeled into the throat.

This suggests a simple geometric interpretation. The acoustic horizon at r_H plays the role of an effective black hole horizon for phonons and small disturbances on the brane, while the throat at radius a plays the role of a minimal surface in the bulk geometry. The flow from r_H down to a is the hydrodynamic realization of the region between the horizon and the throat in a higher-dimensional brane-bulk picture. The near-linear relation $r_H \sim \kappa GM/c^2$ can then be interpreted as a statement about how the minimal 4D throat radius scales with the mass of the defect in the stiff $n = 5$ vacuum.

In subsequent sections we will see that this structure has direct consequences for null rays and strong-field observables. The same index profile $N(\Phi)$ and inflow $u(r)$ that define the acoustic horizon also control the existence and location of a photon sphere, the effective shadow size, and the onset of deviations from Schwarzschild lensing. In this way the thermodynamically fixed fluid parameters ($n = 5, q = 1$) not only close the 1PN dynamical story but also determine a specific class of horizon and throat geometries that can be probed observationally.

7 Photon Sphere, Lensing, and Strong-Field Observables

The acoustic horizon and throat geometry of Sec. 6 fix the near-field flow and refractive index profiles around a defect in the stiff $n = 5$ vacuum. In this section we study how null rays propagate in this background, focusing on the existence and location of an effective photon sphere and the associated shadow size. We then compare the resulting strong-field lensing behavior with the Schwarzschild case and identify regimes where the toy model is expected to agree with or deviate from GR.

7.1 Null rays in the refractive index profile

In the geometric optics limit, null rays on the brane propagate in the effective optical metric

$$ds_{\text{opt}}^2 = -\frac{c^2}{N^2(r)} dt^2 + N^2(r) (dr^2 + r^2 d\Omega^2), \quad (60)$$

with $N(r) \equiv N[\Phi(r)]$ given by Eq. (32). For a static, spherically symmetric defect the potential depends only on r , and the index profile can be written as

$$N(r) = 1 - \frac{1}{2}(n-1) \frac{\Phi(r)}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad \Phi(r) \simeq -\frac{GM}{r}, \quad (61)$$

in the weak-field regime. For $n = 5$ this gives

$$N(r) \simeq 1 + 2 \frac{GM}{rc^2}, \quad (62)$$

which is the familiar $1/r$ profile from Paper II, now understood as a consequence of the thermodynamic matching.

Restricting attention to equatorial rays ($\theta = \pi/2$), the optical metric implies conserved quantities associated with time translation and axial symmetry:

$$E_{\text{opt}} = \frac{c^2}{N^2(r)} \frac{dt}{d\lambda}, \quad L_{\text{opt}} = N^2(r) r^2 \frac{d\phi}{d\lambda}, \quad (63)$$

where λ is an affine parameter. The null condition, $ds_{\text{opt}}^2 = 0$, then yields an effective radial equation of motion of the form

$$\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r; b) = 0, \quad b \equiv \frac{L_{\text{opt}}}{E_{\text{opt}}/c}, \quad (64)$$

where b is the impact parameter. Eliminating $dt/d\lambda$ and $d\phi/d\lambda$ in favor of E_{opt} , L_{opt} , and b gives

$$V_{\text{eff}}(r; b) = -\frac{E_{\text{opt}}^2}{c^2} \left[1 - \frac{b^2}{N^4(r) r^2} \right]. \quad (65)$$

The turning points and circular photon orbits are therefore controlled by the combination $N^4(r) r^2$.

In practice, the lensing calculations in the `lensing_from_flow.wl` script are implemented in terms of the equivalent ray equation for $\phi(r)$,

$$\frac{d\phi}{dr} = \frac{b}{r \sqrt{N^4(r) r^2 - b^2}}, \quad (66)$$

which is integrated numerically using representative weak-field-matched index profiles with finite-size corrections near the throat. The analytic discussion here is intended to highlight the key structures in the optical profile that control the photon sphere and shadow.

7.2 Photon sphere and shadow size

A photon sphere is defined as a marginally stable circular null orbit. In the effective potential language, this corresponds to a radius r_{ph} satisfying

$$V_{\text{eff}}(r_{\text{ph}}; b_{\text{ph}}) = 0, \quad \frac{\partial V_{\text{eff}}}{\partial r}(r_{\text{ph}}; b_{\text{ph}}) = 0, \quad (67)$$

for some critical impact parameter b_{ph} . Using Eq. (65), these conditions can be expressed more directly in terms of the index profile as

$$b_{\text{ph}}^2 = N^4(r_{\text{ph}}) r_{\text{ph}}^2, \quad \frac{d}{dr} [N^4(r) r^2]_{r=r_{\text{ph}}} = 0. \quad (68)$$

The second condition implies

$$\left. \frac{4N'(r)}{N(r)} + \frac{2}{r} \right|_{r=r_{\text{ph}}} = 0, \quad (69)$$

or equivalently

$$\frac{N'(r_{\text{ph}})}{N(r_{\text{ph}})} = -\frac{1}{2r_{\text{ph}}}. \quad (70)$$

For a simple $1/r$ index profile, $N(r) = 1 + \beta_{\text{idx}} GM/(rc^2)$, this weak-field estimate gives

$$r_{\text{ph}} \simeq \frac{\beta_{\text{idx}} GM}{c^2}, \quad b_{\text{ph}} \simeq 4 r_{\text{ph}}, \quad (71)$$

up to corrections of order $(GM/rc^2)^2$. These coefficients should be read only as order-unity estimates extracted from the weak-field tail; a quantitatively reliable photon sphere requires the full nonperturbative optical profile near the throat.

In the superfluid defect model, the actual index profile deviates from a pure $1/r$ law inside the acoustic horizon due to the nonlinear inflow and finite throat size. The representative profiles explored in `lensing_from_flow.wl` therefore use the correct $n = 5$ weak-field tail together with a smooth finite-size correction near the throat. The results can be summarized as follows:

- A photon-sphere-like orbit exists at a radius

$$r_{\text{ph}} \simeq \kappa_{\text{ph}} \frac{GM}{c^2}, \quad (72)$$

with κ_{ph} a dimensionless constant of order unity in the representative profiles studied here.

- The corresponding critical impact parameter scales as

$$b_{\text{ph}} \simeq \eta_{\text{ph}} \frac{GM}{c^2}, \quad (73)$$

with η_{ph} likewise an order-unity, model-dependent coefficient.

- The shadow radius seen by a distant observer, defined by $b_{\text{sh}} \approx b_{\text{ph}}$, is therefore nearly proportional to M , with the detailed coefficient controlled by the throat radius a , the near-throat optical profile, and the finite-size correction used to regularize the strong-field region.

In other words, the $n = 5$ superfluid vacuum predicts a photon-sphere and shadow structure that is qualitatively similar to, but not exactly the same as, that of a Schwarzschild black hole of the same mass. The deviations are controlled by the same throat impedance and mass-radius scaling discussed in Sec. 6.3 and App. D.

7.3 Comparison with GR and observational windows

The near-linear mass scaling of both the acoustic horizon radius r_H and the photon-sphere radius r_{ph} implies that, for sufficiently large and slowly rotating defects, the superfluid defect model will be indistinguishable from GR in classic weak-field and moderately strong-field tests. In particular:

- The 1PN observables considered in Papers I and II (perihelion precession, light bending, Shapiro delay, gravitational redshift) are exactly matched by construction.
- The existence of an acoustic horizon and a photon sphere ensures that null rays experience an effective shadow region with size $b_{\text{sh}} \propto M$, as in Schwarzschild.
- For impact parameters $b \gg b_{\text{ph}}$, the bending angle is dominated by the same $\mathcal{O}(GM/bc^2)$ terms that control standard weak-field lensing.

Deviations from GR are expected to appear in two closely related regimes:

1. **Near-horizon strong-field lensing.** For impact parameters $b \sim b_{\text{ph}}$, the detailed shape of the index profile inside and just outside the acoustic horizon becomes important. The finite throat size and the transition from 3D to 4D flow introduce corrections to r_{ph} and b_{ph} that are not captured by a pure Schwarzschild metric.
2. **Higher-order (2PN) corrections.** The same fluid features that set the throat impedance and mass-radius scaling also control higher-order terms in the PN expansion, such as Φ^2/c^2 and $v^2\Phi^2/c^4$ corrections. These show up in precision timing, high-compactness orbital motion, and in the detailed shape of the lensing cross section near the shadow edge.

From an observational perspective, the most promising windows for testing these deviations are:

- Very long baseline interferometry images of supermassive compact objects (EHT-style observations), where the measured shadow size and photon-ring structure can be compared with the nearly Schwarzschild but slightly modified predictions of the toy model.
- Precision timing of compact binaries in regimes where 2PN and higher corrections are measurable, especially if independent constraints on the mass-radius relation of the compact object are available.
- Numerical simulations of full superfluid dynamics in the $n = 5$ vacuum, using the PDE schemes discussed elsewhere in this series, to quantify how robust the predicted deviations are under more realistic, time-dependent flows.

In summary, the fixed $n = 5$, $M \propto \rho$ fluid not only reproduces the 1PN dynamics of GR but also predicts a specific pattern of small but potentially observable deviations in strong-field lensing and near-horizon structure. These are natural targets for future observational and numerical tests of the superfluid defect toy universe.

8 Predictions, 2PN-Scale Deviations, and Tests

The previous sections have shown that the stiff $n = 5$, $M \propto \rho$ vacuum reproduces the 1PN dynamics of GR and supports an acoustic horizon and photon sphere with approximately Schwarzschild-like scaling. In this section we summarize the main predictive levers of the model, focusing on finite-size effects, the onset of 2PN-scale deviations, and possible observational and numerical tests.

8.1 Finite-size effects and aspect ratio L/a

In the throat picture of Paper V, each defect is associated with a four-dimensional bridge of minimal radius a and characteristic length L in the bulk. From the brane point of view, the far-field potential is effectively monopolar, while the near-field structure depends on the dimensionless aspect ratio

$$\Lambda \equiv \frac{L}{a}. \quad (74)$$

Paper V's stability analysis for a rounded-funnel throat geometry yields a preferred value $\Lambda_\star \simeq 1.85$, suggesting that astrophysical defects in the stable regime should typically have $\Lambda = \mathcal{O}(1)$ rather than $\Lambda \gg 1$. For $\Lambda \gg 1$ the throat appears long and thin, and the near-field region on the brane resembles a quasi-cylindrical cavity; for $\Lambda \sim \mathcal{O}(1)$ the throat is stubby and the transition from 3D to 4D flow is more abrupt.

The inflow and impedance calculations in `4d_throat_impedance_scaling.wl` and `black_hole_scaling.wl` indicate that the mass M , horizon radius r_H , and throat radius a are related schematically by

$$M \sim \mu a^k, \quad r_H \sim \kappa \frac{GM}{c^2}, \quad (75)$$

with $k \lesssim 1$ and μ, κ dimensionless coefficients that depend on Λ and on the detailed boundary conditions in the bulk. In the present standalone audit bundle, these scripts should be read as phenomenological scaling models anchored to the preferred aspect ratio $\Lambda_\star$, not as first-principles derivations of the full throat microphysics. For an exactly Schwarzschild black hole one would have $k = 1$ and a strictly linear mass-radius relation; in the stiff $n = 5$ vacuum the scaling is close to but not exactly linear, leaving room for small finite-size corrections controlled by Λ .

From the brane perspective, deviations from GR first enter through terms that are sensitive to the finite size of the throat region, such as:

- corrections to the far-field potential at order r^{-3} and higher, associated with the multipole structure of the 4D throat,
- modifications to the index profile $N(r)$ inside and just outside the acoustic horizon due to the bending of flow into the bulk, and
- shifts in the effective horizon and photon-sphere radii as functions of Λ .

These effects are naturally of order

$$\epsilon_{\text{fs}} \sim \left(\frac{a}{r_H} \right)^p, \quad (76)$$

with $p \geq 1$ a model-dependent exponent. Since r_H itself scales with M , Eq. (76) implies that finite-size effects are most important for low-mass, high-compactness objects and become progressively less visible for very massive defects with $r_H \gg a$.

8.2 Regimes of agreement and breakdown

It is useful to organize the parameter space of the model in terms of the dimensionless compactness

$$\mathcal{C} \equiv \frac{GM}{r_H c^2} \sim \frac{1}{\kappa}, \quad (77)$$

and the strength of the potential and velocities probed by a given experiment or observation. Three qualitatively distinct regimes can be identified:

1. Weak-field, slow-motion regime (1PN). Here $|\Phi|/c^2 \ll 1$ and $v^2/c^2 \ll 1$ everywhere of interest. In this regime:

- The hybrid 1PN Lagrangian matches the EIH Lagrangian exactly by construction.
- Finite-size corrections are suppressed by powers of $a/r \ll 1$, and are typically much smaller than the already tiny 2PN corrections predicted by GR.
- All solar-system tests and classic binary-pulsar 1PN observables are reproduced.

In practice this regime covers most of the existing precision tests of GR.

2. Moderately strong-field, near-horizon regime. In this regime the potential reaches $|\Phi|/c^2 \sim \mathcal{O}(0.1)$ and orbital velocities approach $v^2/c^2 \sim \mathcal{O}(0.1)$, but the flow is still predominantly 3D and the throat is not fully resolved. Here:

- The effective index profile $N(r)$ remains close to its Schwarzschild counterpart, and the photon-sphere and shadow sizes obey Eqs. (72) and (73) with order-unity coefficients that vary only mildly across the representative profiles studied here.
- 2PN corrections in the toy model are expected to track those of GR up to corrections of order ϵ_{fs} , making this a natural regime in which to look for small deviations.

Current EHT-style images of supermassive compact objects likely probe this regime.

3. Strong-field, throat-resolving regime. Finally, in the immediate vicinity of the acoustic horizon and inside the throat region, the flow is highly nonlinear and the effective potential and index deviate significantly from the Schwarzschild form:

- The transition from 3D inflow on the brane to 4D flow into the bulk leads to order-unity modifications of $N(r)$ for $r \lesssim \mathcal{O}(a)$.
- Higher-order terms in the PN expansion (2PN and beyond) become large, and the series itself may cease to be a useful organizing tool for the dynamics.
- Detailed predictions in this regime require full numerical solutions of the underlying superfluid PDEs in 3D+1 or 4D+1, rather than a purely PN treatment.

At present, this regime is more relevant for numerical experiments than for direct astrophysical observations, though extreme compact binaries and high-precision gravitational-wave measurements may eventually probe aspects of it.

8.3 Astrophysical and numerical tests

The superfluid defect toy model makes several concrete predictions that distinguish it from GR once one goes beyond the strict 1PN regime. These can be grouped into astrophysical tests and numerical experiments.

Astrophysical tests. Promising observational channels include:

- **Black-hole shadow size and shape.** The near-linear scaling of the shadow radius with mass is shared with Schwarzschild, but the proportionality constant is shifted by finite-size and throat effects. Precision measurements of shadow size and photon-ring structure across a range of masses could constrain κ_{ph} and hence Λ .
- **Strong-field lensing profiles.** Detailed measurements of lensing cross sections and caustic structures near the shadow edge could reveal deviations from the Schwarzschild bending angle at the few-percent level, particularly if combined with independent mass measurements.
- **High-compactness binaries and 2PN timing.** In compact binaries where 2PN terms are observationally relevant (for example, in pulsar timing or extreme mass-ratio inspirals), the modified 2PN coefficients of the toy model could manifest as small discrepancies in periastron advance, time delay, or waveform phasing relative to GR predictions.

Numerical experiments. On the modeling side, the most direct way to test and refine the predictions of the superfluid defect model is through full PDE simulations:

- **3D superfluid simulations of single defects.** Solving the nonlinear superfluid equations in 3D around a single throat with $n = 5$ would allow a direct measurement of the inflow profile $u(r)$, the acoustic horizon radius r_H , and the index profile $N(r)$, providing an independent check of the semi-analytic and 1D radial approximations used here.
- **Binary and multi-defect configurations.** Extending these simulations to binary or multi-defect systems would permit a direct comparison between the emergent dynamics and the EIH predictions, testing the robustness of the 1PN matching and probing the onset of 2PN-scale deviations.
- **Ray tracing in simulated flows.** Given a numerically obtained flow and index profile, one can perform ray tracing to compute shadows, photon rings, and lensing patterns, and compare them directly with GR predictions and observations.

Together, these astrophysical and numerical tests provide a roadmap for falsifying or refining the superfluid defect toy model. The key point is that, once the fluid is fixed by the 1PN matching to have $n = 5$ and $M \propto \rho$, the strong-field and 2PN-scale behavior is no longer a free parameter: it is dictated by the same thermodynamic and geometric structure that underlies the success of the model at 1PN.

9 Discussion and Outlook

The superfluid defect toy universe was originally motivated by a simple question: can a single structured vacuum medium reproduce the familiar gravitational and electromagnetic phenomena of general relativity and Maxwell theory, at least in the weak-field regime? Papers I–V answered this question in the affirmative at first post-Newtonian order by constructing an explicit model in which a compressible superfluid vacuum, threaded by defect throats, reproduces Newtonian gravity, 1PN orbital dynamics, light bending, Shapiro delay, Lense–Thirring precession, and an effective electromagnetic sector. However, that success came with several apparent tunings and open questions: the special role of the stiff polytrope $n = 5$, the calibrated wake–mixing ratio

$\alpha^2 = 3/4$ required by the 1PN N -body (EIH) vector sector, and the status of horizons and strong-field structure in the throat picture.

The present paper has addressed these issues from a thermodynamic and geometric perspective. By combining a polytropic equation of state, $P \propto \rho^n$, with a simple equivalence relation between defect mass and vacuum density, $M \propto \rho^q$, and by tracking both scalar (density starvation) and vector (flow and refractive index) responses of the medium, we showed that the requirement of 1PN consistency with the Einstein–Infeld–Hoffmann dynamics acts as a uniqueness condition on the fluid:

- The polytropic index is fixed to $n = 5$; the vacuum must be stiff in precisely the way anticipated by the optics paper.
- The defect mass must track the local vacuum density linearly, $q = 1$, so that $M \propto \rho$.

Within the class of models considered here, there is no alternative choice of (n, q) that reproduces both Newtonian gravity and the full EIH interaction at 1PN order. What appeared in earlier work as a collection of successful but somewhat isolated tunings can thus be understood as the emergent consequence of a single thermodynamic hypothesis and a single equivalence assumption.

This hybrid analysis also clarifies how the wake sector interfaces with the thermodynamic and optical sectors. In the updated Paper III formulation, the completed wake basis yields a positive-definite quadratic wake functional, and EIH matching fixes the wake-mixing ratio $\alpha^2 = 3/4$. In the present paper we treat this α^2 as a calibrated input and focus on what can be learned by combining it with the thermodynamic/equivalence structure that fixes (n, q) .

On the geometric side, we have shown that the same stiff fluid which supports the 1PN dynamics also admits acoustic-horizon and photon-sphere-like structures in the representative strong-field models studied here. Steady inflow in the $n = 5$ vacuum can be organized into a smooth transonic profile with an acoustic horizon at $r = r_H$ where the radial flow speed equals the local sound speed. In the associated phenomenological inflow/impedance fits, the horizon scale remains approximately linear in the mass, $r_H \sim \kappa GM/c^2$, and lies just outside the minimal throat radius a . The corresponding weak-field-matched optical profiles support photon-sphere-like orbits at $r_{\text{ph}} \sim \kappa_{\text{ph}} GM/c^2$ and nearly Schwarzschild shadows, with the detailed coefficients controlled by the throat impedance and aspect ratio L/a .

Taken together, these results complete the conceptual arc of the 1PN program:

- The vacuum is not just any compressible medium; it is uniquely pinned to a stiff $n = 5$ polytrope with $M \propto \rho$ by 1PN consistency.
- The wake/vector sector is consistent with a positive-definite wake functional and a calibrated wake-mixing ratio $\alpha^2 = 3/4$ inherited from Paper III.
- The same fixed fluid determines a specific class of horizon and throat geometries that are observationally accessible through strong-field lensing and shadow measurements.

Limitations and open questions. Despite these advances, several important limitations remain. First, our analysis is restricted to the 1PN expansion plus semi-analytic arguments about the near-horizon flow. We have not derived the full second post-Newtonian (2PN) Lagrangian from the superfluid model, nor have we included radiation reaction or gravitational waves. While preliminary symbolic calculations suggest that the $n = 5$ fluid does not immediately contradict GR at 2PN order, a systematic matching program analogous to the EIH analysis remains to be carried out.

Second, many of our strong-field conclusions rely on reduced models of the flow (steady, spherically symmetric inflow, effective 1D radial equations) and on numerical solutions of those reduced equations. A fully dynamical, three-dimensional simulation of the $n = 5$ superfluid, with realistic initial conditions and defect motions, is needed to validate the existence and robustness of the acoustic horizon, the precise scaling of r_H and r_{ph} , and the stability of the throat under time-dependent perturbations.

Third, the electromagnetic sector, while conceptually unified with gravity through the throat geometry, has been treated largely in the linearized, long-wavelength limit. The interplay between strong gravitational fields, strong electromagnetic fields, and the underlying superfluid dynamics remains to be explored in detail, especially in regimes relevant to compact object magnetospheres and high-energy astrophysical phenomena.

Directions for future work. The natural next steps in this program can be grouped into three broad categories:

1. Toward a full 2PN and radiation theory. A logical continuation of the present work is to extend the hybrid scalar-vector analysis to second post-Newtonian order and beyond. This would involve:

- computing the full 2PN expansion of the hybrid Lagrangian, including all Φ^2/c^2 , $v^2\Phi^2/c^4$, and multi-body cross terms,
- matching these coefficients to the standard 2PN and 2.5PN GR results, or identifying controlled deviations where the toy model predicts a different structure, and
- deriving the effective stress-energy carried by superfluid waves and vortices to understand gravitational radiation and its back reaction on defects.

Such a program would clarify whether the superfluid defect universe can faithfully reproduce the rich phenomenology of binary inspirals and gravitational waves, or whether it predicts specific 2PN-scale departures that could be constrained by current and future observations.

2. Fully nonlinear numerical simulations. On the computational side, a major goal is to move from reduced radial models to full 3D (or 4D) simulations of the superfluid equations. With modern GPU hardware and adaptive mesh refinement, it should be possible to:

- simulate single and binary throats in the $n = 5$ vacuum, resolving both the near-throat and far-field regions,
- measure the emergent mass-radius relation, horizon structure, and index profile directly from the flow, and
- perform ray tracing and wave propagation in the numerically obtained backgrounds, providing parameter-free predictions for shadows, photon rings, and lensing.

Such simulations would also test the stability of the throats and the self-healing properties of the vacuum under strong perturbations.

3. Phenomenology and observational confrontations. Finally, the model invites direct comparison with astrophysical data. Key targets include:

- black-hole imaging (shadow size and shape, photon-ring structure) in systems where independent mass estimates are available,
- precision timing of compact binaries in regimes where 2PN and higher corrections are observable, and
- potential signatures in high-energy emission or polarization patterns arising from the unified superfluid description of gravity and electromagnetism.

In each case, the stiff $n = 5$ fluid and $M \propto \rho$ relation lead to concrete, quantitative predictions once the throat parameters are fixed.

In summary, this paper has shown that the 1PN success of the superfluid defect toy universe is not the result of arbitrary tuning but flows from a tightly constrained thermodynamic and geometric structure. The vacuum is forced into a specific corner of theory space, characterized by a stiff $n = 5$ equation of state, a linear mass-density relation, and a calibrated wake-mixing ratio $\alpha^2 = 3/4$ in the vector sector. This same structure determines the existence and properties of acoustic horizons and photon spheres, providing clear targets for future numerical and observational tests. Whether the model ultimately survives these tests or is ruled out, it offers a concrete example of how a self-healing vacuum might mimic GR to high accuracy at 1PN order while leaving its own fingerprints in the strong-field and 2PN regimes.

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A 1PN Expansion and EIH Matching Details

In this appendix we summarize the algebraic steps that connect the hybrid scalar+vector Lagrangian of Sec. 5 to the Einstein-Infeld-Hoffmann (EIH) coefficients at first post-Newtonian order. The goal is not to reproduce the full multi-body derivation of Paper III, but to make explicit how the polytropic index n and the mass-density exponent q enter the 1PN coefficients, and how the matching conditions select the unique solution $(n, q) = (5, 1)$.

A.1 Setup and series expansion

We begin from the schematic hybrid test-body Lagrangian for a particle moving in a static potential $\Phi(\mathbf{r})$,

$$L_{\text{hyb}} = -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}} + L_{\text{flow}}[\Phi, \mathbf{v}], \quad (78)$$

where:

- $M(\Phi)$ is the effective, potential dependent mass obtained from density starvation,
- $c_s(\Phi)$ is the potential dependent sound speed arising from the polytropic equation of state, and
- L_{flow} encodes the interaction of the defect with the background flow and is responsible for the gravitomagnetic and multi-body vector terms.

In the weak field, slow motion regime we expand

$$M(\Phi) = M_0 \left[1 + q \frac{\Phi}{c^2} + m_2 \frac{\Phi^2}{c^4} + \mathcal{O}\left(\frac{\Phi^3}{c^6}\right) \right], \quad (79)$$

$$\frac{c_s^2(\Phi)}{c^2} = 1 + (n-1) \frac{\Phi}{c^2} + s_2 \frac{\Phi^2}{c^4} + \mathcal{O}\left(\frac{\Phi^3}{c^6}\right), \quad (80)$$

where m_2 and s_2 are coefficients that depend on the detailed microphysics but do not affect the 1PN matching. The squared speed is the only place where n enters the kinematics at this order.

We also define the small expansion parameters

$$\varepsilon_v \equiv \frac{v^2}{c^2}, \quad \varepsilon_\Phi \equiv \frac{|\Phi|}{c^2}, \quad (81)$$

and treat $\varepsilon_v \sim \varepsilon_\Phi \equiv \varepsilon \ll 1$. Terms of order ε are Newtonian, and terms of order ε^2 are 1PN.

A.2 Single body 1PN expansion

For the scalar and kinematic parts of the hybrid Lagrangian we write

$$L_{\text{sc+kin}} \equiv -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}}, \quad (82)$$

and expand the square root using

$$\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3), \quad x \equiv \frac{v^2}{c_s^2(\Phi)}. \quad (83)$$

To 1PN order in ε we keep terms up to x^2 and up to first order in Φ/c^2 inside x . Using Eqs. (79) and (80) we have

$$\frac{v^2}{c_s^2(\Phi)} = \frac{v^2}{c^2} \left[1 - (n-1) \frac{\Phi}{c^2} + \mathcal{O}(\varepsilon^2) \right] = \varepsilon_v - (n-1) \varepsilon_v \varepsilon_\Phi + \mathcal{O}(\varepsilon^3), \quad (84)$$

so that

$$x^2 = \varepsilon_v^2 + \mathcal{O}(\varepsilon^3), \quad (85)$$

and

$$\sqrt{1 - \frac{v^2}{c_s^2(\Phi)}} = 1 - \frac{1}{2}\varepsilon_v + \frac{1}{2}(n-1) \varepsilon_v \varepsilon_\Phi - \frac{1}{8}\varepsilon_v^2 + \mathcal{O}(\varepsilon^3). \quad (86)$$

Multiplying Eqs. (79) and (86), and discarding terms of order ε^3 , gives

$$\begin{aligned} L_{\text{sc+kin}} &= -M_0 c^2 [1 + q \varepsilon_\Phi] \left[1 - \frac{1}{2} \varepsilon_v + \frac{1}{2} (n-1) \varepsilon_v \varepsilon_\Phi - \frac{1}{8} \varepsilon_v^2 \right] + \mathcal{O}(\varepsilon^3) \\ &= -M_0 c^2 - M_0 q \Phi + \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} + \frac{1}{2} M_0 [q - (n-1)] \frac{\Phi v^2}{c^2} + \mathcal{O}(\varepsilon^3), \end{aligned} \quad (87)$$

which reproduces Eq. (45) when the constant rest energy $-M_0 c^2$ is dropped. The key points are:

- The Newtonian kinetic term $+\frac{1}{2} M_0 v^2$ and the 1PN kinematic correction $+\frac{1}{8} M_0 v^4/c^2$ are independent of n and q .
- The Newtonian potential energy $-M_0 q \Phi$ depends only on q .
- The mixed 1PN term $\propto \Phi v^2/c^2$ depends on the combination $\frac{1}{2}[q - (n-1)]$, which we identified as $C_{\text{hyb}}(n, q)$ in Eq. (46).

A.3 EIH template in the test-mass limit

For comparison, consider a nonspinning test mass M_0 moving in the static field of a much heavier source of mass M . In the standard EIH gauge the 1PN Lagrangian for the test mass can be written schematically as

$$L_{\text{EIH}} = \frac{1}{2} M_0 v^2 - M_0 \Phi + A_{\text{EIH}} M_0 \frac{v^4}{c^2} + B_{\text{EIH}} M_0 \frac{\Phi v^2}{c^2} + D_{\text{EIH}} M_0 \frac{\Phi^2}{c^2} + \dots, \quad (88)$$

where $\Phi = -GM/r$, and A_{EIH} , B_{EIH} , and D_{EIH} are numerical coefficients fixed by GR. The dots denote higher order PN terms and, in the multi-body case, additional velocity dependent interactions between bodies.

The important part for the present analysis is that:

- the Newtonian terms have fixed coefficients $+\frac{1}{2}$ and -1 in front of $M_0 v^2$ and $M_0 \Phi$, and
- the 1PN mixed term has a fixed coefficient B_{EIH} .

In particular, the GR value of B_{EIH} is gauge invariant within the class of isotropic coordinate systems used in the previous papers, and is the same coefficient that appears in the two-body EIH Lagrangian for the pairwise terms of the form $\Phi_{AB} v_A^2$.

A.4 Matching conditions and solution for (n, q)

Comparing Eqs. (87) and (88), and recalling that L_{flow} does not contribute to the single-body Newtonian terms, we obtain two algebraic matching conditions at 1PN order:

$$(\text{Newtonian potential}) : \quad -M_0 q \Phi = -M_0 \Phi \quad \Rightarrow \quad q = 1, \quad (89)$$

$$(\text{mixed 1PN term}) : \quad C_{\text{hyb}}(n, q) M_0 \frac{\Phi v^2}{c^2} + C_{\text{flow}}(n) M_0 \frac{\Phi v^2}{c^2} = B_{\text{EIH}} M_0 \frac{\Phi v^2}{c^2}, \quad (90)$$

where we have written the total mixed 1PN coefficient as the sum of the hybrid scalar+kinematic contribution $C_{\text{hyb}}(n, q)$ from Eq. (46) and a pure flow contribution $C_{\text{flow}}(n)$ arising from integrating out the vector degrees of freedom in L_{flow} .

The Newtonian matching condition (89) fixes

$$q = 1, \quad (91)$$

as in Sec. 5.2. With this choice, the mixed 1PN coefficient becomes

$$C_{\text{hyb}}(n, q = 1) = \frac{2 - n}{2}, \quad (92)$$

and Eq. (90) can be rewritten as

$$\frac{2 - n}{2} + C_{\text{flow}}(n) = B_{\text{EIH}}. \quad (93)$$

The compressible flow kernel that enters L_{flow} is the same one used in Paper III to parameterize the longitudinal and transverse response in terms of α . There, the EIH matching conditions for the multi-body velocity dependent terms led to an algebraic relation between α and the fluid parameters that can be expressed as

$$C_{\text{flow}}(n) = f_{\text{flow}}(n), \quad (94)$$

for some smooth function f_{flow} determined by the choice of equation of state and by the longitudinal/transverse mixing.

In practice, the matching is carried out as follows:

1. Construct the full two-body hybrid Lagrangian, including both scalar+kinematic and flow contributions, as a function of (n, q) .
2. Expand this Lagrangian to 1PN order using symbolic manipulations (for example, in Mathematica; see the `fundamental_fluid_check.wl` and `density_cross_talk_derivation.wl` scripts).
3. Collect the coefficients of the independent 1PN structures: v_A^4 , v_B^4 , $v_A^2 \Phi_B$, $\mathbf{v}_A \cdot \mathbf{v}_B$, $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$, and $\Phi_A \Phi_B$.
4. Equate these coefficients to the corresponding EIH coefficients obtained from GR.

The resulting system of algebraic equations is overdetermined: there are more coefficient conditions than free parameters. Solving this system symbolically yields a unique real solution for (n, q) :

$$n = 5, \quad q = 1. \quad (95)$$

In particular, the equation (93) for the mixed $\Phi v^2/c^2$ term is satisfied only at $n = 5$, because the flow contribution $C_{\text{flow}}(n)$ that reproduces the remaining EIH vector coefficients intersects the GR value B_{EIH} at that single point.

A.5 Summary

To summarize, the key outcomes of the 1PN expansion and EIH matching are:

- The Newtonian limit fixes the mass-density exponent to $q = 1$, ensuring that defect mass is proportional to the local vacuum density.

- The mixed 1PN coefficient $\propto \Phi v^2/c^2$ depends on the combination

$$C_{\text{hyb}}(n, q) + C_{\text{flow}}(n),$$

where $C_{\text{hyb}}(n, q)$ comes from scalar and kinematic effects and $C_{\text{flow}}(n)$ comes from the vector flow.

- Matching this combination, together with the remaining 1PN coefficients, to the GR EIH values leads to a unique polytropic index $n = 5$.

Thus, within the class of polytropic superfluid vacua with power-law mass-density relations, the requirement of 1PN consistency with GR selects a single fluid: a stiff $n = 5$ vacuum in which defect masses track the local density, $M \propto \rho$.

B Scalar–Vector Decomposition and Coefficient Tables

In this appendix we make explicit which pieces of the 1PN coefficients come from the scalar (density starvation / variable mass) sector and which come from the vector (sound speed / flow) sector. The goal is to clarify how the hybrid scalar+vector dynamics reproduces the Einstein–Infeld–Hoffmann (EIH) coefficients, and how the genuinely wake-mediated vector pieces may be packaged into the calibrated wake-mixing ratio $\alpha^2 = 3/4$ inherited from Paper III.

Throughout, we work in the two-body case with labels A and B and relative separation $\mathbf{r}_{AB}$, and we restrict attention to the velocity dependent terms at 1PN order. The full symbolic derivations are contained in the Mathematica scripts `fundamental_fluid.check.wl` and `density_cross_talk.derivation.wl`; here we summarize the structure of the results.

B.1 Scalar contributions

For a single body in a static potential, the scalar sector (variable mass with fixed propagation speed) contributes the Lagrangian

$$L_{\text{sc}} = -M(\Phi) c^2 \sqrt{1 - \frac{v^2}{c^2}}, \quad M(\Phi) = M_0 \left(1 + q \frac{\Phi}{c^2} \right) + \mathcal{O}\left(\frac{\Phi^2}{c^4}\right), \quad (96)$$

whose 1PN expansion is

$$L_{\text{sc}} = \frac{1}{2} M_0 v^2 - M_0 q \Phi + \frac{1}{8} M_0 \frac{v^4}{c^2} + \frac{1}{2} M_0 q \frac{\Phi v^2}{c^2} + \mathcal{O}\left(\frac{\Phi^2}{c^2}, \frac{\Phi v^4}{c^4}\right), \quad (97)$$

as in Eq. (23).

In a two-body system, the potential felt by body A is $\Phi_A = -GM_B/r_{AB}$ and vice versa. To leading PN order, the scalar sector contributes the following terms to the pairwise Lagrangian:

$$L_{\text{sc}}^{(AB)} = -q \frac{GM_A M_B}{r_{AB}} + \frac{1}{2} q \frac{GM_A M_B}{r_{AB} c^2} (v_A^2 + v_B^2) + \cdots, \quad (98)$$

where the ellipsis denotes terms that are higher order in the PN expansion or independent of q . There is no scalar-sector contribution to the purely vector structures $\mathbf{v}_A \cdot \mathbf{v}_B$ or $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$ at this order.

We can summarize the scalar contributions to the 1PN pairwise coefficients in the schematic EIH form

$$L_{AB}^{(1\text{PN})} = \frac{GM_A M_B}{r_{AB} c^2} [C_1(v_A^2 + v_B^2) + C_2 \mathbf{v}_A \cdot \mathbf{v}_B + C_3 (\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})] + \dots, \quad (99)$$

as

$$C_1^{(\text{sc})} = \frac{1}{2}q, \quad C_2^{(\text{sc})} = 0, \quad C_3^{(\text{sc})} = 0. \quad (100)$$

B.2 Vector contributions

The vector sector includes both the dependence of the sound speed on the potential and the flow-mediated interactions. The kinematic piece with variable sound speed but fixed mass has Lagrangian

$$L_{\text{kin,vec}} = -M_0 c^2 \sqrt{1 - \frac{v^2}{c_s^2(\Phi)}}, \quad \frac{c_s^2(\Phi)}{c^2} = 1 + (n-1) \frac{\Phi}{c^2} + \dots, \quad (101)$$

with 1PN expansion

$$L_{\text{kin,vec}} = \frac{1}{2} M_0 v^2 + \frac{1}{8} M_0 \frac{v^4}{c^2} - \frac{1}{2} (n-1) M_0 \frac{\Phi v^2}{c^2} + \dots, \quad (102)$$

as in Eq. (39). In a two-body system this yields kinematic contributions to C_1 of the form

$$C_1^{(\text{kin})} = -\frac{1}{2}(n-1). \quad (103)$$

The flow-induced vector potential L_{flow} encodes the overlap of the velocity fields sourced by different defects. In the far field, and for slowly moving bodies, the resulting pairwise vector interaction can be written in the same coefficient form as Eq. (99) with

$$C_1^{(\text{flow})}(n), \quad C_2^{(\text{flow})}(n), \quad C_3^{(\text{flow})}(n), \quad (104)$$

treated as functions of the polytropic index n . These functions are determined by the compressible flow kernel and the longitudinal/transverse mixing parameter α introduced in Paper III.

The detailed form of $C_i^{(\text{flow})}(n)$ is somewhat involved and was obtained symbolically using the Mathematica scripts; here it is sufficient to note that:

- the same flow kernel reproduces the correct GR coefficients for the purely vector structures $\mathbf{v}_A \cdot \mathbf{v}_B$ and $(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$ when expressed in terms of α , and
- the dependence of $C_1^{(\text{flow})}(n)$ on n is such that EIH matching yields a unique solution at $n = 5$.

Summarizing, the vector contributions decompose schematically as

$$C_1^{(\text{vec})}(n) = C_1^{(\text{kin})}(n) + C_1^{(\text{flow})}(n) = -\frac{1}{2}(n-1) + C_1^{(\text{flow})}(n), \quad (105)$$

$$C_2^{(\text{vec})}(n) = C_2^{(\text{flow})}(n), \quad (106)$$

$$C_3^{(\text{vec})}(n) = C_3^{(\text{flow})}(n). \quad (107)$$

B.3 Combined coefficients and comparison with GR

Combining scalar and vector sectors, the total coefficients in Eq. (99) take the form

$$\begin{aligned} C_1^{(\text{tot})}(n, q) &= C_1^{(\text{sc})}(q) + C_1^{(\text{vec})}(n) \\ &= \frac{1}{2}q - \frac{1}{2}(n-1) + C_1^{(\text{flow})}(n), \end{aligned} \quad (108)$$

$$C_2^{(\text{tot})}(n) = C_2^{(\text{vec})}(n) = C_2^{(\text{flow})}(n), \quad (109)$$

$$C_3^{(\text{tot})}(n) = C_3^{(\text{vec})}(n) = C_3^{(\text{flow})}(n). \quad (110)$$

The Einstein–Infeld–Hoffmann Lagrangian for two nonspinning point masses has fixed GR coefficients

$$C_1^{(\text{GR})}, \quad C_2^{(\text{GR})}, \quad C_3^{(\text{GR})}, \quad (111)$$

which can be read off from the standard EIH expressions.

We can summarize the matching problem in the table below:

term	scalar	vector (kin + flow)	GR target
$v_A^2 + v_B^2$	$C_1^{(\text{sc})} = \frac{1}{2}q$	$C_1^{(\text{vec})} = -\frac{1}{2}(n-1) + C_1^{(\text{flow})}(n)$	$C_1^{(\text{GR})}$
$\mathbf{v}_A \cdot \mathbf{v}_B$	$C_2^{(\text{sc})} = 0$	$C_2^{(\text{vec})} = C_2^{(\text{flow})}(n)$	$C_2^{(\text{GR})}$
$(\mathbf{v}_A \cdot \mathbf{n}_{AB})(\mathbf{v}_B \cdot \mathbf{n}_{AB})$	$C_3^{(\text{sc})} = 0$	$C_3^{(\text{vec})} = C_3^{(\text{flow})}(n)$	$C_3^{(\text{GR})}$

Imposing the Newtonian limit $q = 1$ fixes the scalar column. The remaining matching conditions are then

$$C_1^{(\text{tot})}(n, q = 1) = C_1^{(\text{GR})}, \quad (112)$$

$$C_2^{(\text{tot})}(n) = C_2^{(\text{GR})}, \quad (113)$$

$$C_3^{(\text{tot})}(n) = C_3^{(\text{GR})}. \quad (114)$$

Solving these equations with the explicit $C_i^{(\text{flow})}(n)$ obtained from the compressible flow kernel yields a unique real solution

$$n = 5, \quad q = 1. \quad (115)$$

Any other choice of (n, q) fails to reproduce the GR values of the velocity dependent coefficients.

B.4 Relation to the α parameter

Paper III expressed the long-range multi-body wake interaction in terms of a longitudinal/transverse decomposition with a single mixing parameter α . In that language the wake energy may be written schematically as

$$E_{\text{hydro}} = E_{\perp} + \alpha^2 E_{\parallel}, \quad (116)$$

where $E_{\perp}$ and $E_{\parallel}$ denote transverse and longitudinal contributions constructed from the completed wake basis. The updated EIH matching fixes a positive wake balance,

$$\alpha^2 = \frac{3}{4}, \quad (117)$$

which may be interpreted as the calibrated ratio of longitudinal to transverse wake power required for the 1PN vector sector.

The scalar–vector decomposition used in the present paper does not attempt to rederive this wake kernel. Instead, it clarifies how the thermodynamic inputs (n, q) determine the *scalar* and *index/kinematic* pieces of the 1PN coefficients, while the genuinely wake–mediated vector pieces can be packaged into the single parameter α^2 . The unique solution $(n, q) = (5, 1)$ thus should be read as a statement about the vacuum microphysics consistent with both the optical sector and the EIH matching, with $\alpha^2 = 3/4$ supplied as an inherited calibration from Paper III.

C Nonlinear Inflow and Transonic Solution

This appendix collects the details of the nonlinear inflow model used to define the acoustic horizon in Sec. 6. We work with a steady, spherically symmetric accretion flow in the stiff $n = 5$ vacuum, and we show how the transonic condition arises at the level of the governing ordinary differential equation (ODE). For the present audit bundle, the accompanying `nonlinear_profile_solver.wl` script implements a reduced profile model that enforces the same continuity, sonic, and far–field constraints without attempting a full Paper V throat solve.

C.1 Governing equations

We consider a steady flow of the superfluid vacuum toward a throat of radius a . The flow is taken to be spherically symmetric on the brane, with radial velocity $u(r) < 0$ (inward) and density $\rho(r)$. The basic equations are:

Mass conservation. A constant mass flux $\dot{M}_{\text{flux}}$ implies

$$\dot{M}_{\text{flux}} = 4\pi r^2 \rho(r) u(r) = \text{const}, \quad (118)$$

which can be solved for $\rho(r)$ as

$$\rho(r) = \frac{\dot{M}_{\text{flux}}}{4\pi r^2 u(r)}. \quad (119)$$

Momentum balance (Euler equation). In the absence of viscosity, the radial Euler equation reads

$$u(r) \frac{du}{dr} = -\frac{1}{\rho(r)} \frac{dP}{dr} - \frac{d\Phi_{\text{eff}}}{dr}, \quad (120)$$

where $P(\rho)$ is the pressure and $\Phi_{\text{eff}}(r)$ is an effective potential that includes both the long–range monopolar contribution $-GM/r$ and near–throat corrections due to the 3D–to–4D transition. We parametrize

$$\Phi_{\text{eff}}(r) = -\frac{GM}{r} + \Phi_{\text{throat}}(r; a), \quad (121)$$

where $\Phi_{\text{throat}}(r; a)$ is a short–range potential that decays rapidly for $r \gg a$ and encodes the finite throat size. Its specific form is not needed for the PN analysis, but is fixed in the `nonlinear_profile_solver.wl` audit model by a smooth near–throat regularization chosen to keep the profile transonic and weak–field normalized.

Equation of state. The stiff superfluid vacuum is modeled as a polytrope

$$P(\rho) = K\rho^n, \quad n = 5, \quad (122)$$

with sound speed

$$c_s^2(\rho) = \frac{dP}{d\rho} = Kn\rho^{n-1}. \quad (123)$$

The background density ρ_0 and the constant K are chosen so that $c_s(\rho_0) = c$ far from the throat.

C.2 Reduction to a single ODE

Using the barotropic relation $dP/dr = c_s^2(\rho) d\rho/dr$ and eliminating $d\rho/dr$ in favor of du/dr via Eq. (119), we obtain a first-order ODE for $u(r)$. Differentiating Eq. (119) gives

$$\frac{d\rho}{dr} = \rho(r) \left[-\frac{2}{r} - \frac{1}{u(r)} \frac{du}{dr} \right]. \quad (124)$$

Substituting into dP/dr and then into Eq. (120) yields

$$u \frac{du}{dr} = -c_s^2(\rho) \left[-\frac{2}{r} - \frac{1}{u} \frac{du}{dr} \right] - \frac{d\Phi_{\text{eff}}}{dr}, \quad (125)$$

or, after rearranging,

$$(u^2 - c_s^2) \frac{du}{dr} = u \left[\frac{2c_s^2}{r} - \frac{d\Phi_{\text{eff}}}{dr} \right]. \quad (126)$$

Comparing with Eq. (55), we see that the throat effects enter through the effective potential:

$$\mathcal{F}_{\text{throat}}(r; a, n) \equiv -\frac{d\Phi_{\text{throat}}}{dr}, \quad (127)$$

so that Eq. (126) can be written as

$$(u^2 - c_s^2) \frac{du}{dr} = u \left[\frac{2c_s^2}{r} - \frac{d}{dr} \left(-\frac{GM}{r} \right) + \mathcal{F}_{\text{throat}}(r; a, n) \right], \quad (128)$$

which is exactly Eq. (55) in the main text.

It is often convenient to recast this ODE in terms of the Mach number $\mathcal{M}(r)$,

$$\mathcal{M}(r) \equiv \frac{|u(r)|}{c_s(\rho(r))}, \quad (129)$$

and a dimensionless radius $x \equiv r/a$. In these variables, the ODE takes the generic form

$$(\mathcal{M}^2 - 1) \frac{d \log \mathcal{M}}{d \log x} = \mathcal{G}(x; \mathcal{M}, \Lambda), \quad \Lambda \equiv \frac{L}{a}, \quad (130)$$

where $\mathcal{G}$ is a dimensionless function that encodes the detailed dependence of the effective potential and sound speed on x , and Λ is the throat aspect ratio introduced in Eq. (74). The explicit expression for $\mathcal{G}(x; \mathcal{M}, \Lambda)$ is model dependent; the reduced profile ansatz used in the standalone audit script is recorded in the comments of `nonlinear_profile_solver.wl`.

C.3 Sonic point and regularity condition

The structure of Eq. (126) (or equivalently Eq. (130)) implies the existence of a critical (sonic) point where $u^2 = c_s^2$ or $\mathcal{M} = 1$. At such a point the coefficient of du/dr (or $d\mathcal{M}/dx$) in the ODE vanishes, and a smooth solution requires the right-hand side to vanish as well. This yields the transonic conditions

$$u^2(r_H) = c_s^2(r_H), \quad \left. \frac{2c_s^2(r_H)}{r_H} - \frac{d\Phi_{\text{eff}}}{dr} \right|_{r_H} = 0, \quad (131)$$

which coincide with Eq. (56) when Φ_{eff} is split as in Eq. (121). In dimensionless form this becomes

$$\mathcal{M}(x_H) = 1, \quad \mathcal{G}(x_H; \mathcal{M} = 1, \Lambda) = 0, \quad (132)$$

where $x_H \equiv r_H/a$.

The sonic point is thus determined by a pair of conditions:

- the Mach number equals unity, $\mathcal{M} = 1$, and
- the effective driving term in the ODE vanishes at the same radius.

Physically, the first condition expresses the equality of advection speed and signal speed, while the second expresses a balance between pressure gradient, gravity, and throat-induced forces at the sonic point.

C.4 Boundary conditions and numerical method

The inflow ODE is a two-point boundary value problem. The natural boundary conditions are:

- **Outer boundary.** At large radii $r \gg r_H$, the flow approaches a subsonic, nearly homogeneous background:

$$u(r) \rightarrow u_\infty, \quad \rho(r) \rightarrow \rho_0, \quad c_s(r) \rightarrow c,$$

with $|u_\infty| \ll c$. In practice, the numerical solution is initialized at a finite outer radius r_{max} with $\mathcal{M}(r_{\text{max}}) \ll 1$.

- **Inner boundary.** At the throat radius $r \approx a$, the flow must match the imposed throat boundary condition. This is encoded in the choice of $\Phi_{\text{throat}}(r; a)$ and in the requirement that the mass flux $\dot{M}_{\text{flux}}$ be finite.

The `nonlinear_profile_solver.wl` audit script does not solve the full ODE directly. Instead, it constructs a representative profile by imposing:

1. a chosen sonic radius r_H and depleted horizon density $\rho(r_H) = \rho_H < \rho_0$,
2. the sonic matching condition $|u(r_H)| = c_s(r_H)$,
3. the continuity relation $4\pi r^2 \rho(r) u(r) = \text{const}$, and
4. a smooth interpolation to $(u, \rho) \rightarrow (0, \rho_0)$ in the far field.

This reduced construction is sufficient for the present paper's internal consistency checks: it exhibits a smooth transonic crossing, it respects the $n = 5$ EOS, and it provides a representative profile for the strong-field optical estimates discussed below.

C.5 Asymptotic behavior

It is useful to record the leading asymptotic behavior of the solution in the regimes $r \gg r_H$ and $r \approx r_H$.

Far field, $r \gg r_H$. In the far field, the flow is weak and the density is close to ρ_0 , so $c_s \approx c$ and $|u| \ll c$. The ODE simplifies to

$$u \frac{du}{dr} \approx -\frac{d\Phi_{\text{eff}}}{dr} \approx -\frac{d}{dr} \left(-\frac{GM}{r} \right), \quad (133)$$

which integrates to

$$\frac{1}{2}u^2(r) \approx -\Phi(r) + \text{const} = \frac{GM}{r} + \text{const}. \quad (134)$$

With the boundary condition $u \rightarrow u_\infty$ as $r \rightarrow \infty$, the far-field velocity behaves as

$$u(r) \approx u_\infty - \sqrt{\frac{2GM}{r}} + \dots, \quad (135)$$

showing the expected acceleration into the potential well.

Near sonic point, $r \approx r_H$. Expanding the ODE (126) around $r = r_H$ with $u^2(r_H) = c_s^2(r_H)$ and using the transonic conditions (131), one finds that the derivative du/dr remains finite,

$$\left. \frac{du}{dr} \right|_{r_H} = \frac{u}{2c_s^2} \left[\frac{2c_s^2}{r} - \frac{d\Phi_{\text{eff}}}{dr} + \frac{\partial(c_s^2)}{\partial r} \right] \Big|_{r=r_H}, \quad (136)$$

where the derivative of c_s^2 with respect to r is computed via $dc_s^2/dr = (dc_s^2/d\rho)(d\rho/dr)$ and $d\rho/dr$ is obtained from mass conservation. This regularity is the hallmark of a physically acceptable transonic solution and is enforced in the numerical scheme by imposing the sonic conditions at r_H .

C.6 Summary

The key points of this appendix are:

- Steady, spherically symmetric inflow in the $n = 5$ vacuum reduces to a single nonlinear ODE for the radial velocity $u(r)$, or equivalently for the Mach number $\mathcal{M}(r)$.
- This ODE has a critical point where $u^2 = c_s^2$, and a smooth solution requires the transonic conditions (131) to hold at a radius r_H .
- For physically relevant boundary conditions, there is a unique transonic solution in which the flow is subsonic at large radii and supersonic near the throat, with $r_H \gtrsim a$.

These properties underlie the definition of the acoustic horizon in Sec. 6 and feed directly into the index profiles used in the lensing calculations of Sec. 7.

D Throat Impedance and Mass–Radius Scaling

This appendix summarizes the scaling arguments and numerical checks that relate the throat radius a , the acoustic horizon radius r_H , and the defect mass M in the stiff $n = 5$ superfluid vacuum. The goal is to clarify why the model predicts a nearly linear mass–radius relation, $r_H \propto M$, and how small deviations from exact linearity arise through the finite impedance of the throat.

The calculations described here are implemented in the Mathematica scripts `4d_throat_impedance_scaling.wl` and `black_hole_scaling.wl`. In the present audit bundle these scripts should be read as phenomenological scaling models anchored to the preferred aspect ratio $\Lambda_\star$; they illustrate how finite throat impedance can keep the effective exponent close to unity, rather than deriving that behavior microscopically from a full bulk solve.

D.1 Throat impedance and mass flux

In the throat picture of Paper V, each defect is associated with a 4D bridge of minimal radius a that connects the 3D brane to the bulk superfluid. The throat acts as a conduit for mass flux from the brane into the bulk. On the brane, this appears as an effective sink with mass M ; in the bulk, it appears as a flux tube of cross sectional area πa^2 .

The steady inflow solution of App. C implies a constant mass flux from the brane into the throat,

$$\dot{M}_{\text{flux}} = 4\pi r^2 \rho(r) u(r), \quad (137)$$

which must match the flux through the throat in the bulk. The effective impedance of the throat can be defined as

$$Z_{\text{th}} \equiv \frac{\Delta P}{\dot{M}_{\text{flux}}}, \quad (138)$$

where ΔP is the pressure drop between the brane side and the bulk side of the throat. For a linear, quasi–static response one expects $\Delta P \sim \rho_0 c^2$ and a flux scaling like $\dot{M}_{\text{flux}} \sim \rho_0 c a^2$, so that

$$Z_{\text{th}} \sim \frac{\rho_0 c^2}{\rho_0 c a^2} \sim \frac{c}{a^2}. \quad (139)$$

More generally, the detailed 4D geometry modifies the prefactor and introduces a weak dependence on the aspect ratio $\Lambda = L/a$, but the dominant scaling with a is expected to be $Z_{\text{th}} \propto a^{-2}$.

The defect mass M is controlled by how much of the vacuum is drained and stored in the throat and bulk. A simple dimensional estimate, confirmed by the representative scaling model in `black_hole_scaling.wl`, is

$$M \sim \rho_0 a^3 \mathcal{F}(\Lambda), \quad (140)$$

where $\mathcal{F}(\Lambda)$ is a dimensionless function that encodes the effect of the throat length and bulk boundary conditions. For a perfectly spherical 3D defect one would have $\mathcal{F} \sim \mathcal{O}(1)$ and a strictly cubic scaling $M \propto a^3$; for a 4D throat embedded in a 3D brane, the effective exponent is renormalized by the way flux lines spread into the bulk.

D.2 Effective mass–radius exponent

To quantify this renormalization, the `4d_throat_impedance_scaling.wl` script evaluates a local effective exponent k_{eff} defined by

$$k \equiv \frac{d \log M}{d \log a}, \quad (141)$$

for representative finite-impedance fits of the relation between M and a . Operationally, one tracks how the local power-law slope approaches unity as a increases.

The main findings can be summarized as:

- For a wide range of aspect ratios Λ and boundary conditions, the effective exponent k lies in a narrow band around unity,

$$k \approx 1 \pm \delta_k, \quad \delta_k \ll 1, \quad (142)$$

with the precise value of δ_k depending weakly on Λ .

- In the limit of very long throats ($\Lambda \gg 1$), the exponent approaches $k \rightarrow 1$ from above or below, depending on the bulk geometry and the imposed outer boundary conditions in the bulk.
- For $\Lambda \sim \mathcal{O}(1)$, modest deviations from unity appear, reflecting the fact that the brane sees a compact, stubby object whose effective volume and cross section do not scale as a simple power of a .

These results motivate the phenomenological scaling

$$M \simeq \mu_* a^{k_*}, \quad k_* \lesssim 1, \quad (143)$$

with μ_* and k_* determined by the microphysics of the throat and by the global topology of the bulk. The near-linear scaling $k_* \approx 1$ is what enables the model to mimic the Schwarzschild relation $r_s = 2GM/c^2$ over many orders of magnitude while still allowing room for small finite-size corrections.

D.3 Horizon radius and compactness

The acoustic horizon radius r_H is determined by the transonic condition (131), which involves both the gravitational potential and the sound speed. For the $n = 5$ vacuum, the combined inflow-impedance fits in `black_hole_scaling.wl` yield a relation of the form

$$r_H \simeq \kappa_* \frac{GM}{c^2} \simeq \kappa_* \frac{G\mu_*}{c^2} a^{k_*}, \quad (144)$$

where κ_* is a dimensionless constant of order unity. For $k_* = 1$ this reduces to a strictly linear relation between r_H and M ; for $k_* \neq 1$ the scaling is only approximately linear over a finite range.

The dimensionless compactness parameter $\mathcal{C} = GM/(r_H c^2)$ defined in Eq. (77) is then

$$\mathcal{C} \simeq \frac{1}{\kappa_*} a^{1-k_*}. \quad (145)$$

For $k_* = 1$ the compactness is constant and the horizon radius is exactly proportional to the mass, as in Schwarzschild. For $k_* \neq 1$, the compactness acquires a mild dependence on a , which can be interpreted as a finite-size correction to the Schwarzschild relation.

Numerical scans in `black_hole_scaling.wl` support the following qualitative picture:

- For large defects (large a and M), the effective exponent k_* approaches unity and the compactness approaches a constant, so the model is effectively indistinguishable from GR at the level of the mass-radius relation and horizon scaling.
- For smaller defects, or for throats with $\Lambda \sim \mathcal{O}(1)$, the deviations from $k_* = 1$ become more pronounced, leading to small but potentially observable departures from the Schwarzschild compactness and shadow size.

D.4 Implications for strong-field observables

The near-linear mass-radius scaling has direct implications for the strong-field observables discussed in Sec. 7. In particular:

- The photon sphere radius r_{ph} inherits the same approximate linear scaling with M ,

$$r_{\text{ph}} \simeq \kappa_{\text{ph}}(a) \frac{GM}{c^2},$$

with $\kappa_{\text{ph}}(a)$ varying only weakly with a when $k_* \approx 1$.

- The critical impact parameter b_{ph} and hence the shadow radius likewise scale nearly linearly with M , with small corrections governed by the same deviation of k_* from unity.
- The finite-size parameter ϵ_{fs} introduced in Eq. (76) can be expressed in terms of the departure of k_* from unity and the ratio a/r_H , providing a convenient way to parameterize deviations from Schwarzschild behavior in observational fits.

In summary, the throat impedance and inflow calculations show that, in the stiff $n = 5$ vacuum, the defect mass M , throat radius a , and horizon radius r_H are tied together by an almost linear mass-radius relation. This explains why the superfluid defect model can so closely mimic the Schwarzschild geometry in the strong-field regime while still predicting small, controlled deviations that are natural targets for future numerical and observational tests.

E Lensing from Flow: Numerical Setup

This appendix summarizes the numerical setup used to compute null-ray trajectories, bending angles, and shadow radii in the fixed $n = 5$ superfluid vacuum. The calculations are implemented in the `lensing_from_flow.wl` script, which works with representative weak-field-matched index profiles augmented by a smooth near-throat finite-size correction and outputs ray trajectories and effective photon-sphere parameters.

E.1 Effective index profile from the inflow

The starting point is the spherically symmetric inflow solution $u(r)$, $\rho(r)$ for the $n = 5$ vacuum, obtained as described in App. C. From $\rho(r)$ we compute the local sound speed using the polytropic equation of state

$$c_s^2(r) = \frac{dP}{d\rho} = Kn\rho^{n-1}(r), \quad n = 5, \quad (146)$$

with K chosen so that $c_s(\rho_0) = c$ in the far field. The effective refractive index is then

$$N(r) \equiv \frac{c}{c_s(r)}. \quad (147)$$

In practice, the standalone script works with an analytic profile $N(r)$ chosen so that:

- its weak-field tail matches the $n = 5$ coefficient of Eq. (62), and
- a smooth finite-size correction regularizes the near-throat region.

The radial velocity $u(r)$ enters this audit model only indirectly, through the requirement that the near-throat correction be consistent with an acoustic horizon at $r_H \gtrsim a$. The ray tracing itself is performed entirely in the optical metric determined by $N(r)$.

E.2 Ray equations and impact parameter

On the brane, null rays propagate in the effective optical metric

$$ds_{\text{opt}}^2 = -\frac{c^2}{N^2(r)} dt^2 + N^2(r) (dr^2 + r^2 d\Omega^2), \quad (148)$$

as in Eq. (60). Restricting to the equatorial plane $\theta = \pi/2$, the line element reduces to

$$ds_{\text{opt}}^2 = -\frac{c^2}{N^2(r)} dt^2 + N^2(r) (dr^2 + r^2 d\phi^2). \quad (149)$$

The metric admits two Killing vectors, ∂_t and ∂_ϕ , leading to conserved quantities

$$E_{\text{opt}} = \frac{c^2}{N^2(r)} \frac{dt}{d\lambda}, \quad L_{\text{opt}} = N^2(r) r^2 \frac{d\phi}{d\lambda}, \quad (150)$$

where λ is an affine parameter. The ratio

$$b \equiv \frac{L_{\text{opt}}}{E_{\text{opt}}/c} \quad (151)$$

plays the role of an impact parameter.

The null condition $ds_{\text{opt}}^2 = 0$ yields the radial equation

$$N^2(r) \left(\frac{dr}{d\lambda} \right)^2 + N^2(r) r^2 \left(\frac{d\phi}{d\lambda} \right)^2 = \frac{c^2}{N^2(r)} \left(\frac{dt}{d\lambda} \right)^2, \quad (152)$$

which, after eliminating $dt/d\lambda$ and $d\phi/d\lambda$ in favor of E_{opt} , L_{opt} , and b , can be written as

$$\left(\frac{dr}{d\lambda} \right)^2 = \frac{E_{\text{opt}}^2}{c^2} \left[1 - \frac{b^2}{N^4(r) r^2} \right]. \quad (153)$$

A more convenient form for numerical integration, equivalent to Eq. (66) in the main text, is

$$\frac{d\phi}{dr} = \frac{b}{r \sqrt{N^4(r) r^2 - b^2}}. \quad (154)$$

For a given impact parameter b , this equation describes the angular deflection of a ray as it passes through the index profile $N(r)$.

E.3 Boundary conditions and integration strategy

The bending angle for a ray that comes in from infinity, reaches a minimum radius r_{min} , and returns to infinity is

$$\Delta\phi(b) = 2 \int_{r_{\text{min}}}^{\infty} \frac{b}{r \sqrt{N^4(r) r^2 - b^2}} dr - \pi, \quad (155)$$

where r_{min} is defined implicitly by

$$N^4(r_{\text{min}}) r_{\text{min}}^2 = b^2, \quad (156)$$

so that the denominator in Eq. (154) changes sign. For impact parameters smaller than a critical value b_{ph} , the equation $N^4(r) r^2 = b^2$ has no real solution outside the horizon, and the ray plunges into the throat; this defines the effective shadow.

The numerical integration proceeds as follows:

1. For a given impact parameter b , solve $N^4(r) r^2 = b^2$ for $r_{\min}$ using a root-finding method (for example, Newton-Raphson), starting from an initial guess near the expected turning point.
2. Evaluate the integral in Eq. (155) from $r = r_{\min}$ up to an outer radius $r_{\max}$ where $N(r) \approx 1$ and the contribution to the deflection becomes negligible. The integral is performed using an adaptive quadrature routine, with the integrand regularized near $r_{\min}$ if necessary.
3. Compute $\Delta\phi(b)$ and record whether the ray escapes to infinity or is captured by the throat (no turning point outside r_H).

By scanning over b , one obtains the bending-angle function $\Delta\phi(b)$, the critical impact parameter b_{ph} at which the bending angle diverges (corresponding to the photon sphere), and the range of b for which the ray is captured.

E.4 Photon sphere and shadow extraction

The photon sphere radius r_{ph} and critical impact parameter b_{ph} are determined numerically by the conditions

$$N^4(r_{\text{ph}}) r_{\text{ph}}^2 = b_{\text{ph}}^2, \quad \frac{d}{dr} [N^4(r) r^2]_{r=r_{\text{ph}}} = 0, \quad (157)$$

which are equivalent to Eqs. (68) and (70) in the main text. In practice, the script implements this as a two-parameter root-finding problem in $(r_{\text{ph}}, b_{\text{ph}})$, using the spline-interpolated $N(r)$ and its derivative.

Once b_{ph} is known, the effective shadow radius seen by a distant observer is taken to be

$$b_{\text{sh}} \approx b_{\text{ph}}, \quad (158)$$

with corrections from higher-order terms in the index profile typically smaller than the uncertainties associated with the throat parameters themselves.

The `lensing_from_flow.wl` script outputs:

- The numerically determined r_{ph} and b_{ph} for a given set of throat and inflow parameters.
- The bending-angle function $\Delta\phi(b)$ over a range of impact parameters, which can be compared with the corresponding Schwarzschild result.
- Sample ray trajectories $r(\phi)$ illustrating whether rays with given b escape, orbit near the photon sphere, or are captured by the throat.

E.5 Limitations and approximations

Several simplifying assumptions are built into this numerical setup:

- The flow is assumed to be steady and spherically symmetric on the brane; angular structure and time dependence are neglected.
- The index profile $N(r)$ is taken to be determined entirely by the radial density profile $\rho(r)$ in the $n = 5$ vacuum; any additional anisotropies or frequency dependence of the effective index are ignored.
- The rays are traced in the effective optical metric only on the brane; the full 4D throat geometry and bulk propagation are not modeled explicitly.

Within these approximations, the numerical scheme provides a controlled way to connect the hydrodynamic inflow and throat structure to observable lensing signatures, and to quantify the deviations from Schwarzschild behavior induced by the fixed $n = 5$ superfluid vacuum.

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